

Chapter 6

Time Division Switching

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Outline

- ☐ Introduction
 - ☐ Basic Time Division Space Switching
 - ☐ Basic Time Division Time Switching
 - ☐ Time Multiplexed Space Switching
 - ☐ Time Multiplexed Time Switching
 - ☐ Combination Switching
 - ☐ Three-stage Combination Switching
 - ☐ N-Stage Combination Switching
-

Introduction

□ Review of Space Division Switching

- Single-stage: **Dedicated** switching element for a specific pair of inlet-outlet.
 - Multi-stage: Switching elements **shared** by inlet-outlet pairs.
 - A Common feature: Switching elements involved in a connection serve **only one** speech circuit to pass through continuous speech signal.
-

Introduction

□ Time Division Switching

- Features of PAM/PCM signal
 - Discrete in time domain
 - Transmission of PAM/PCM signal
 - Multiplexing
 - Periodical time slots
 - For PAM/PCM signals, it is possible for a number of active speech circuits to share a single switching element, and therefore reduce the number of switching elements significantly.
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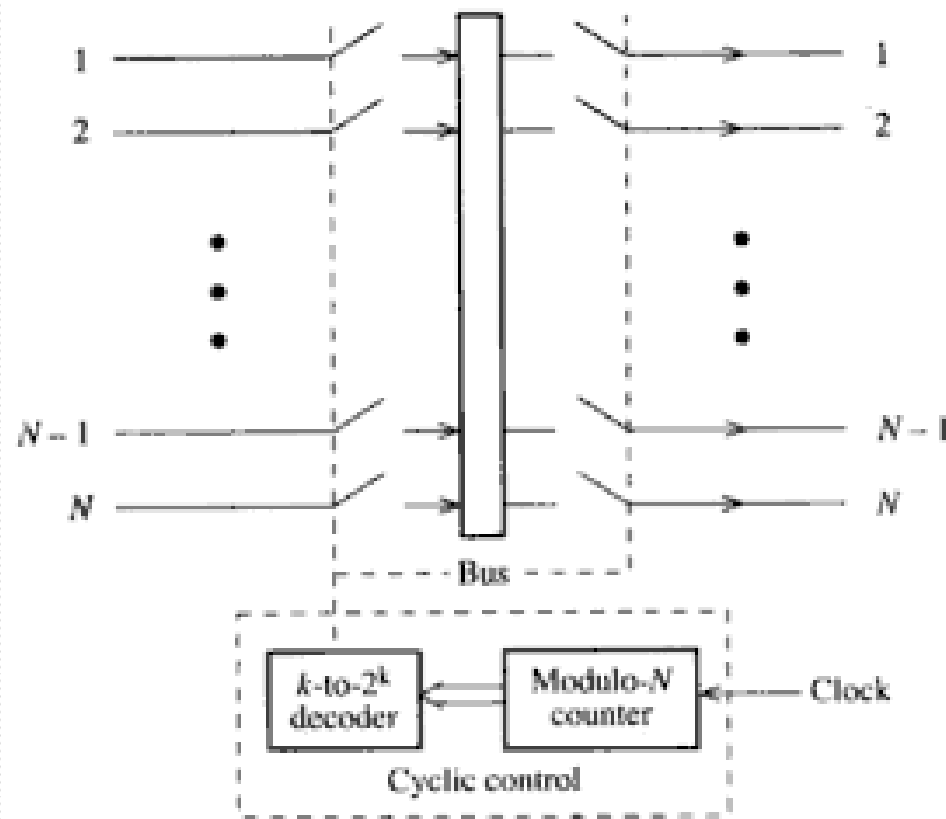
6.1 Basic time division space switching

- NxN time division space switching
 - Architecture
 - Analog time division space switching (PAM)
 - Analog bus is employed
 - Digital time division space switching (PCM)
 - Digital bus is employed
 - Switching capacity (simultaneous conversations supported)

$$SC = 125/t_s$$

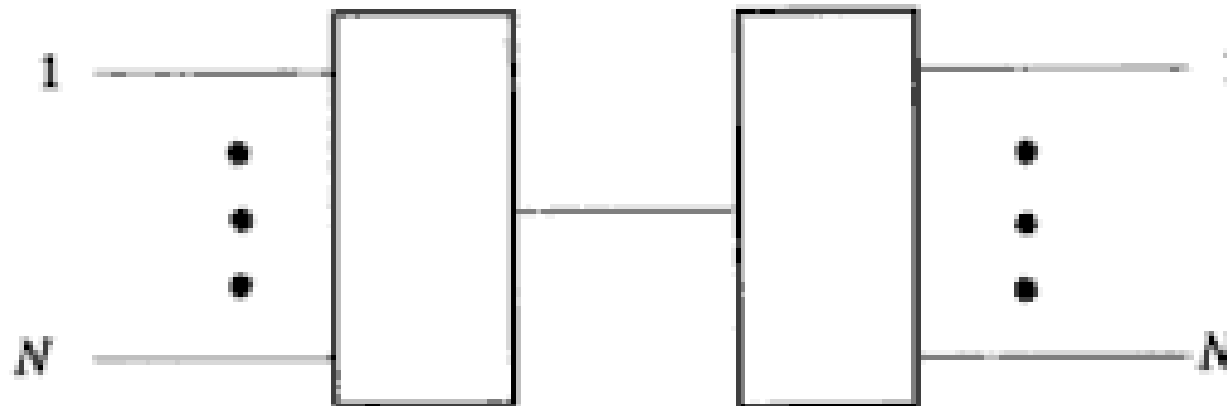
Note: The sampling frequency is 8kHz, the corresponding sampling interval is 125 microseconds, t_s in microsecond is the time to setup a connection and transfer the sample value.

Time division space switching



(a) Switching structure

Time division space switching



(b) Two-stage equivalent.

Fig. 6.1 Simple PAM time division switching.

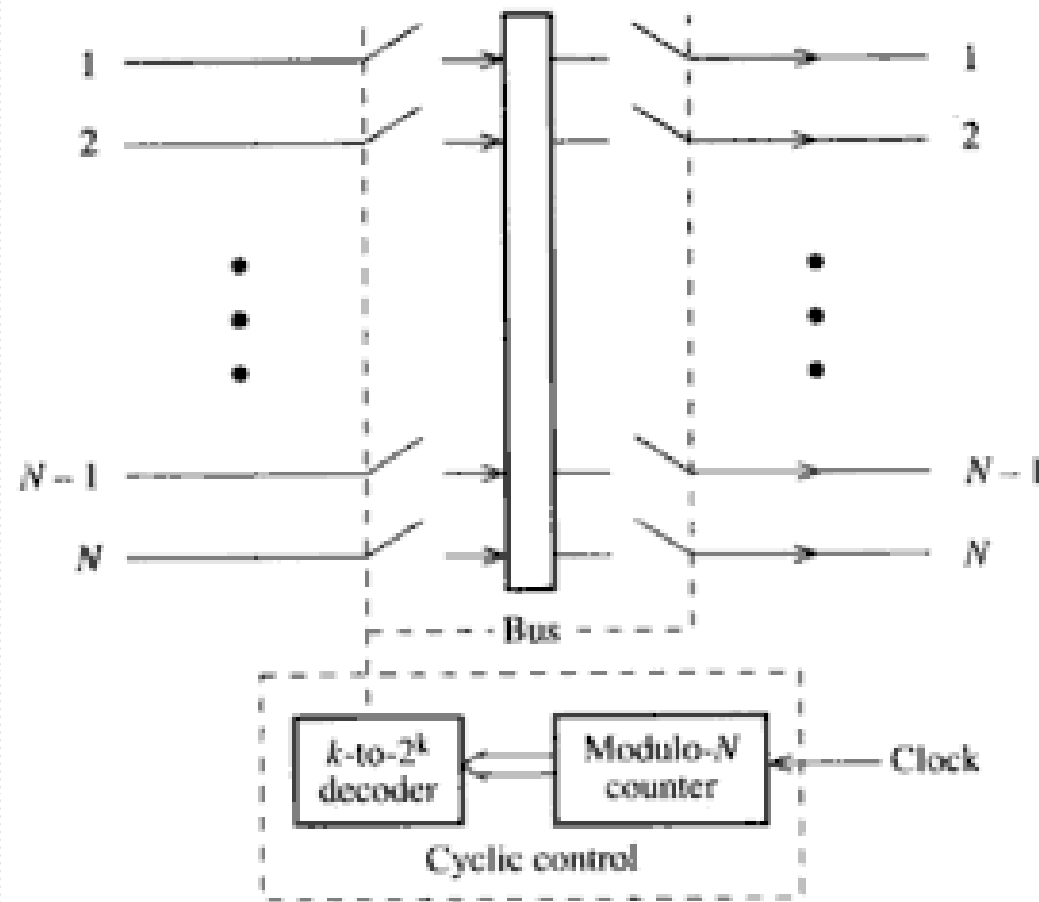
Control mechanisms

- ☐ Cyclic control in synchronism
 - ☐ Input-controlled / Input-driven
 - ☐ Output-controlled / Output-driven
-

Control mechanisms

- Cyclic control in synchronism
 - Cyclically select/scan inlets and outlets
 - Fixed one-to-one correspondence
 - One switching element shared by all connections
 - A modulo-N counter & a k-to- 2^k decoder
$$k = \lceil \log_2 N \rceil$$
 - Remarks:
 - There is no switching.
 - It lacks full availability.
-

Cyclic control in synchronism



nonblocking

Control mechanisms

- Input-controlled / Input-driven
 - The control on the input side works in a cyclic manner.
 - The control on output side is memory-based and changes synchronously with the input side.
 - Full availability is obtained by using the programmable feature of the memory.
-

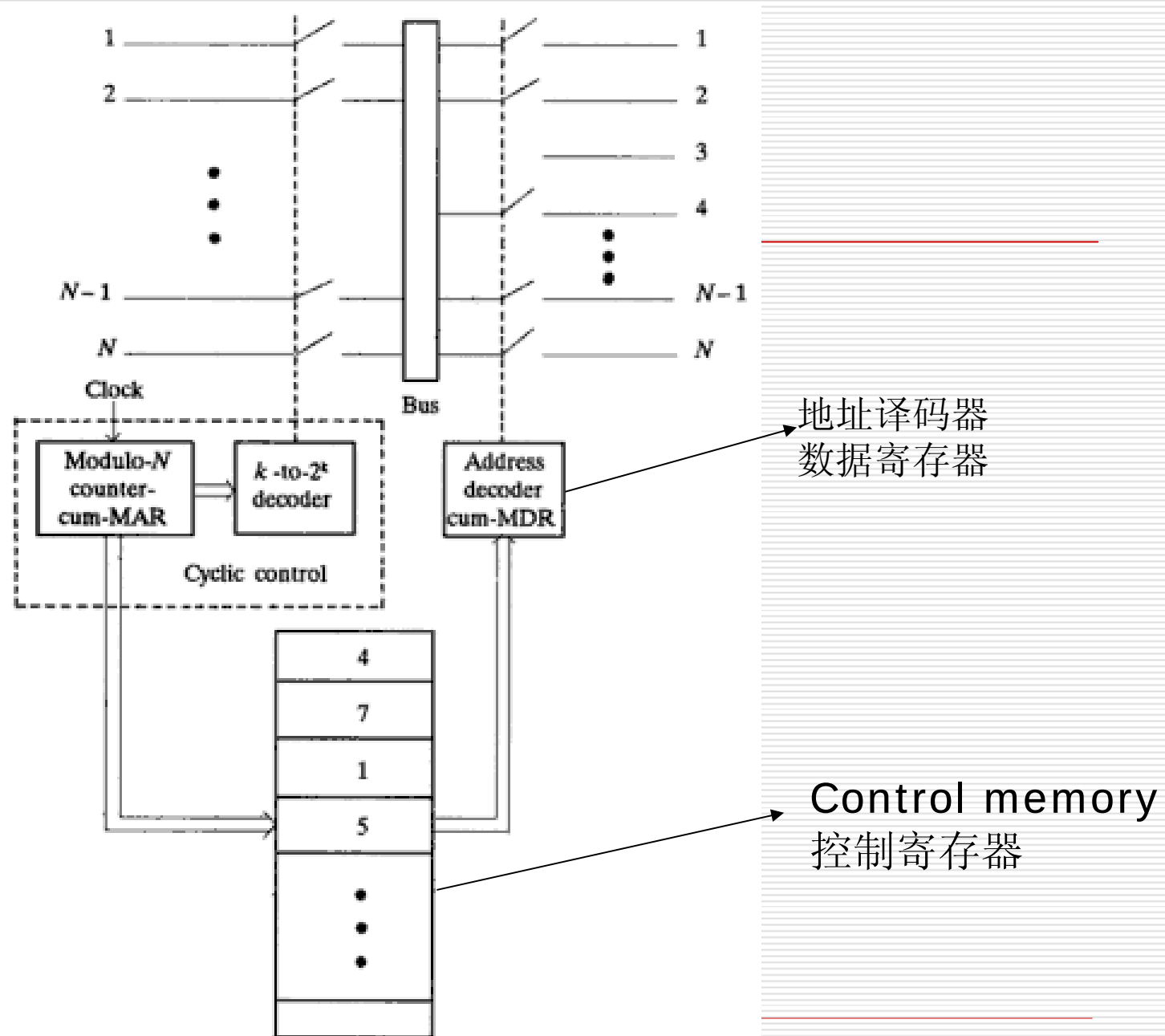


Fig. 6.2 Input-controlled time division space switch.

Control mechanisms

□ Output-controlled

- A **dual** scheme of the input-controlled
 - The control on the output side works in a cyclic manner.
 - **The control on input side is memory-based** and changes synchronously with the output side.
 - **Full availability is obtained** by using the programmable feature of the memory.
 - **Broadcast** is supported.
-

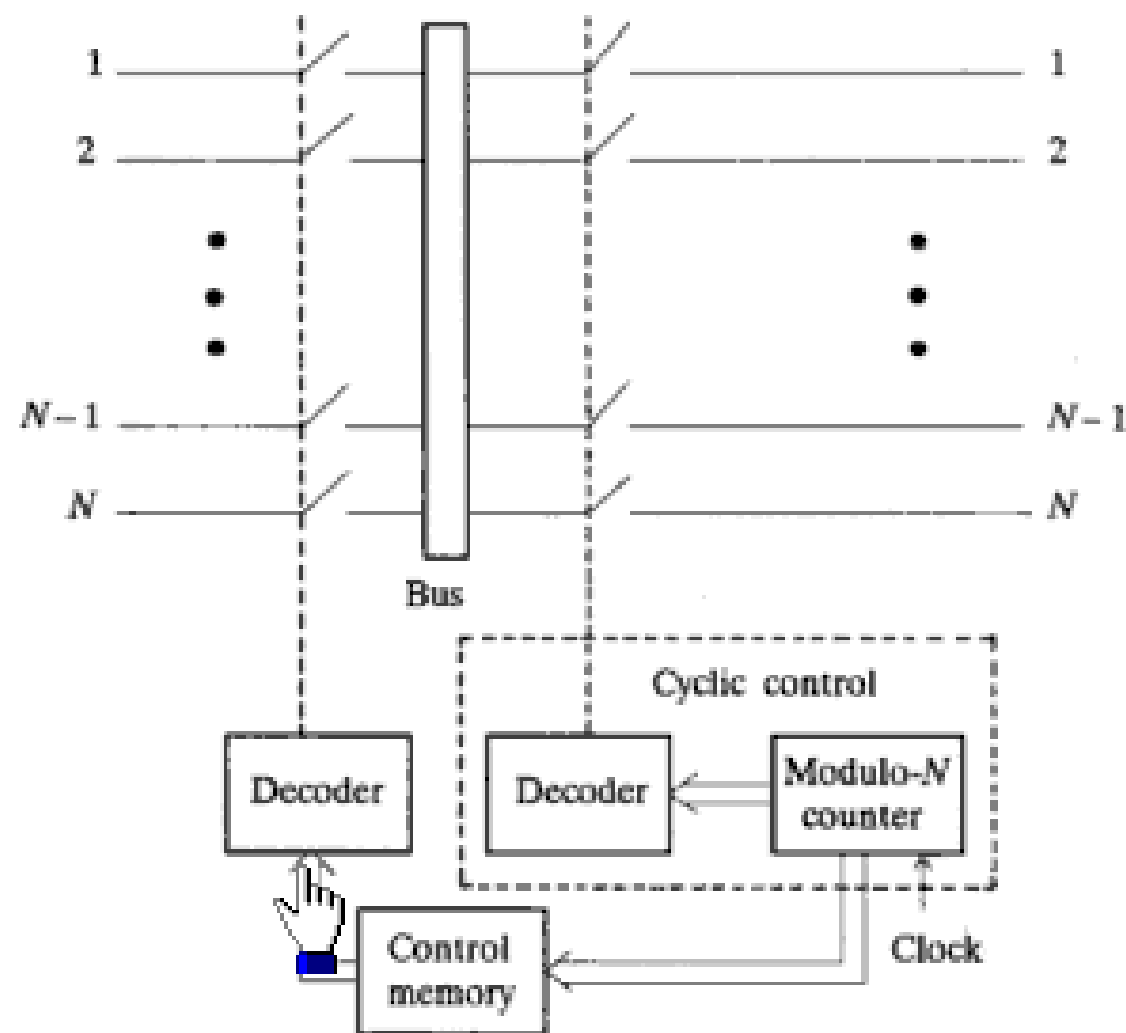


Fig. 6.3 Output-controlled time division space switch.

Remarks on both input-controlled and output-controlled

- The **switching capacity** is equal to the number of inlets or outlets N.

$$N = SC = 125 / (t_i + t_m + t_d + t_t)$$

t_i : time to increment the modulo-N counter

t_m : time to read the control memory

t_d : time to decode address and select inlet/outlet

t_t : time to transfer the sample value from inlet to outlet

- **Clock rate** of the counter

$$\text{Clock rate} = 8N \text{ kHz}$$

Remarks on both input-controlled and output-controlled

☐ Configuration for bi-direction transfer

- Scheme 1: two independent buses used, one for each direction.
- Scheme 2: one bus used, time division multiplexing for two-way transfer.

☐ Folded network

- Both input-controlled and output-controlled can be used to support folded network connections.
-

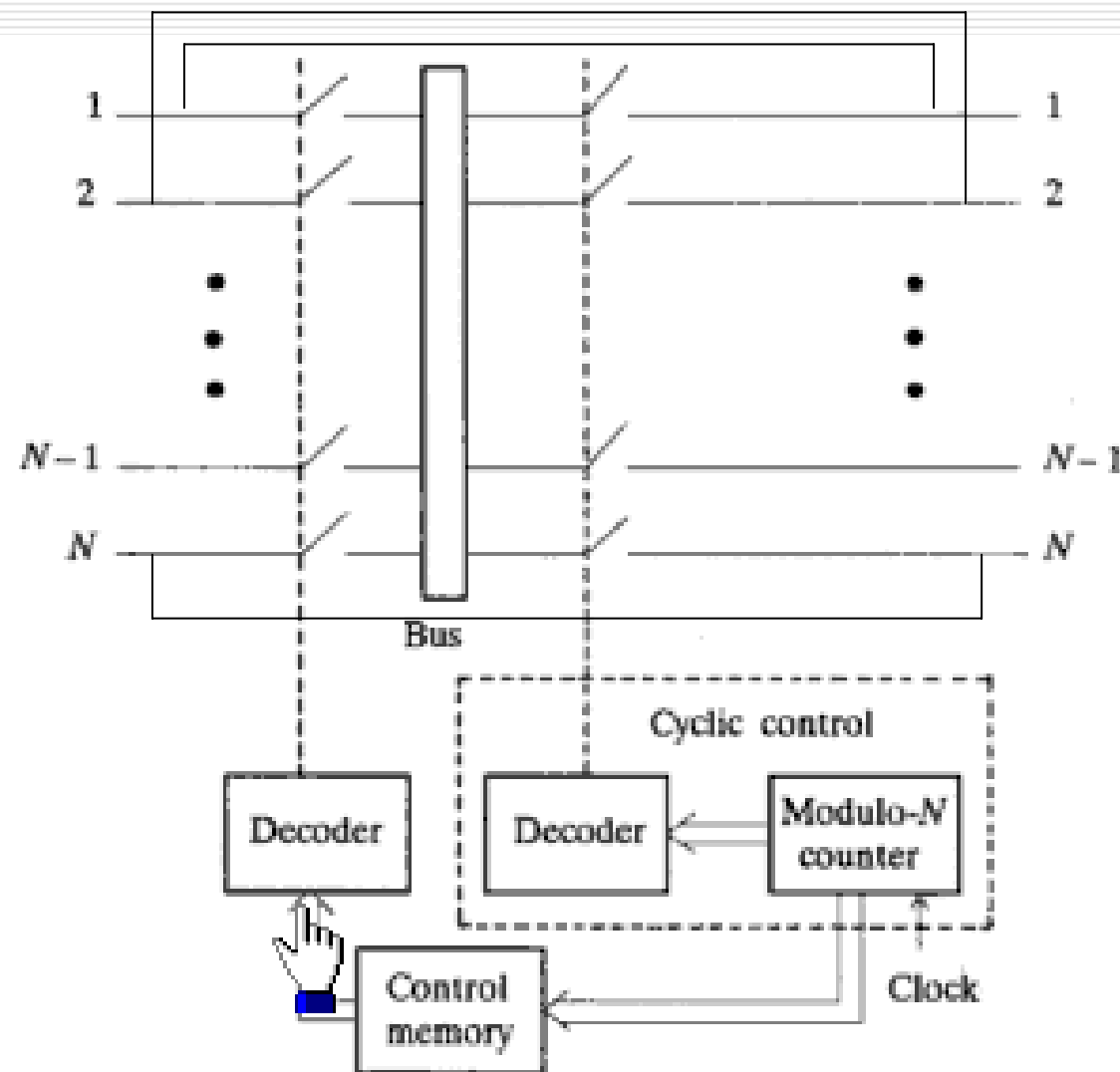


Fig. 6.3 Output-controlled time division space switch.

Remarks on both input-controlled and output-controlled

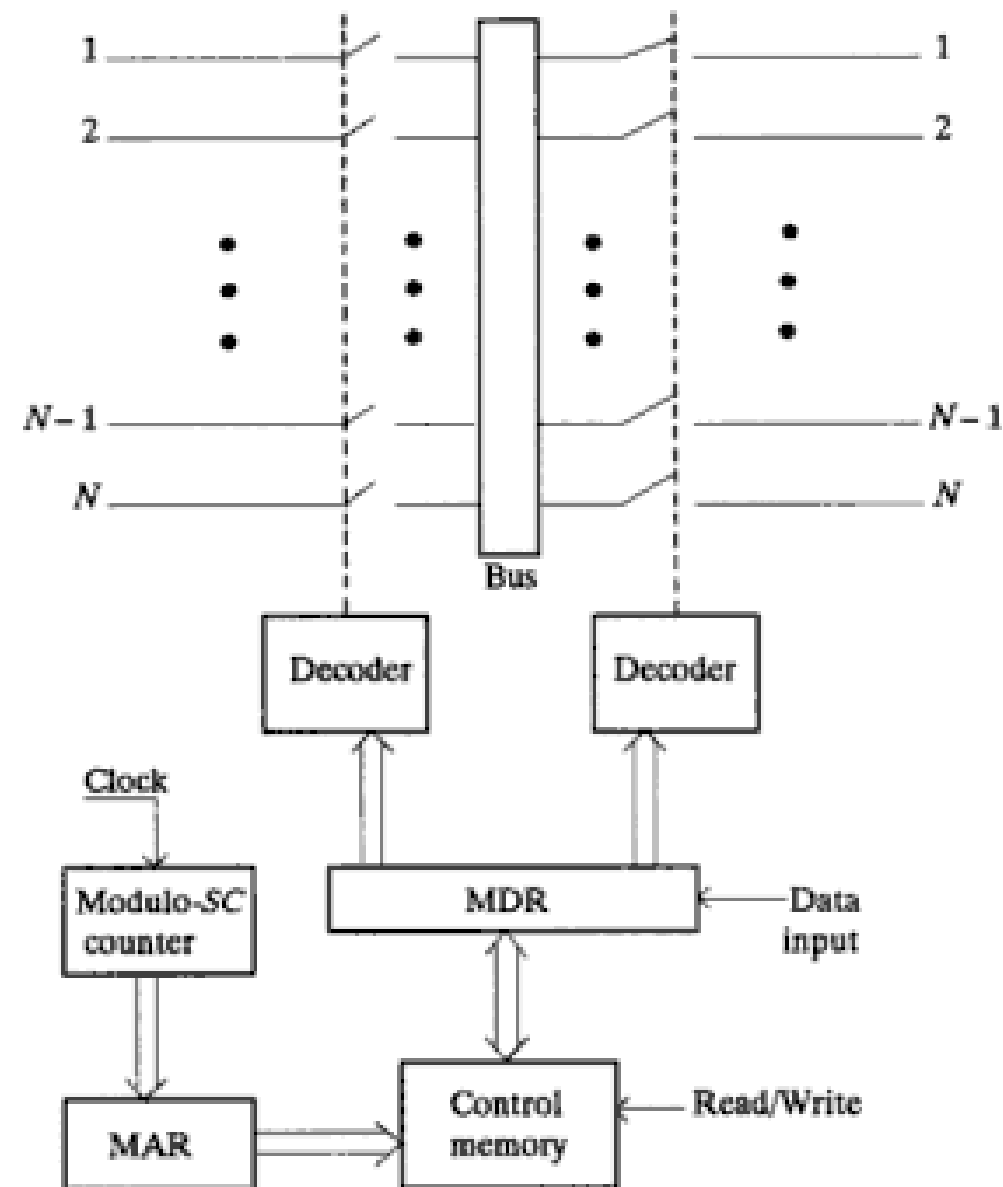
- ❑ **Limitation** on the number of subscribers
 - In both input-controlled and output-controlled, all the lines (inlets or outlets) are scanned whether they are active or not.
 - The number of subscribers is limited by the **time cost of a single speech circuit**.
 - ❑ Nonblocking in nature
-

Parameters of time division space switching network (vs. Space division)

- Number of switching elements
 - $S = N + N = 2N$ ($S = 2N$)
 - Switching capacity
 - $SC = N$ ($SC = 1$)
 - Traffic handling capacity
 - $TC = N/N = 1$ ($TC = 1/N$)
 - Cost of the switching network
 - $C = \text{cost of switching elements and memory}$
 $= 2N + N = 3N$ ($C = 2N$)
 - Cost capacity index
 - $CCI = SC / (C/N) = N/3$ ($CCI = 1 / (2N/N) = 0.5$)
-

Generalized time division space switching

- How to support **more subscribers** in time division space switching?
 - **Memory-controlled** time division space switching (Generalized ~)
 - Both inlets and outlets are controlled by memory.
 - The capacity of control memory used is equal to switching capacity SC.
 - A large number of subscribers **share** the control memory efficiently.
 - **Blocking** probability depends on the traffic characteristics.
-



MDR = memory data address MAR = memory address register

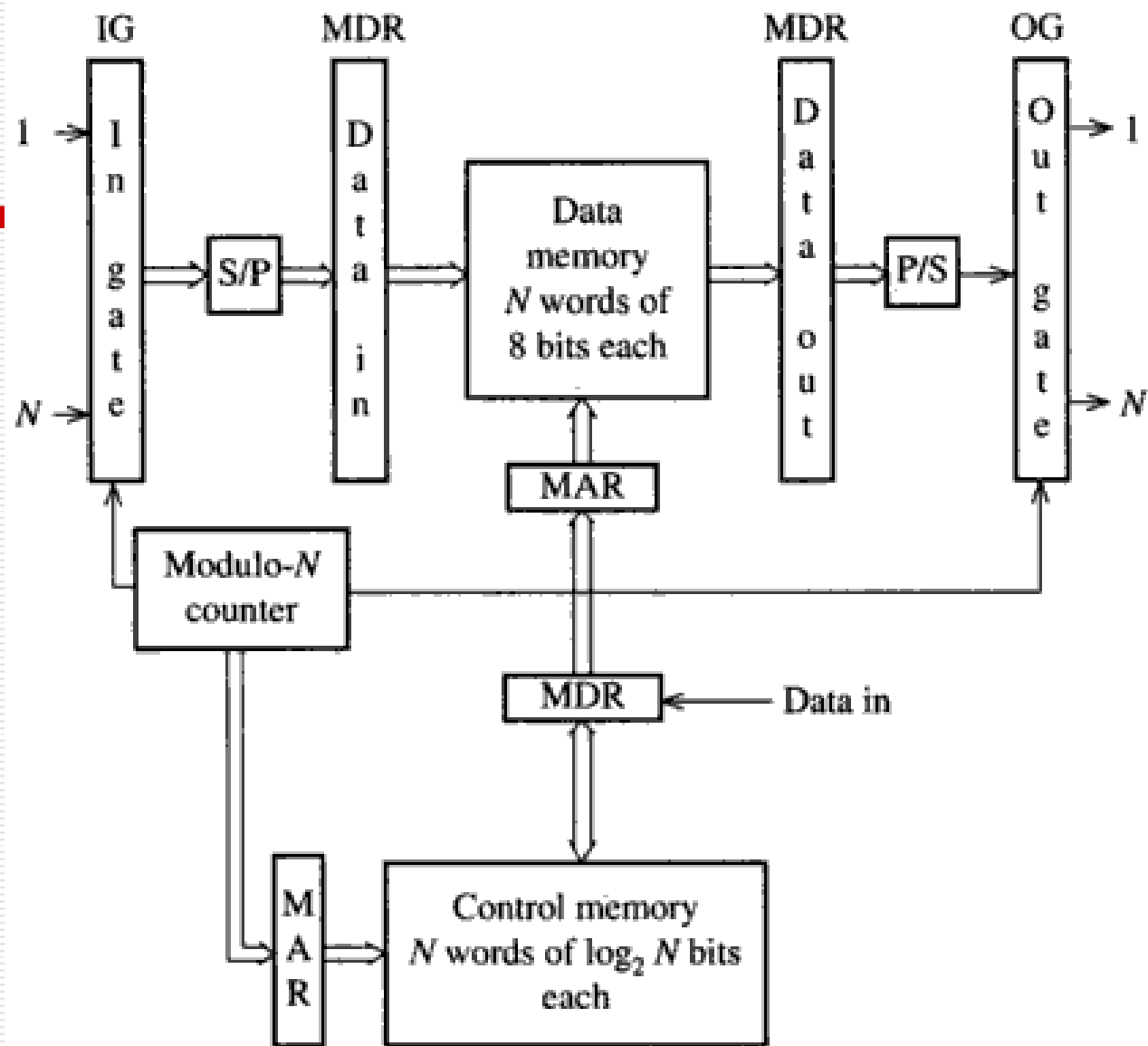
Fig. 6.4 Generalised time division space switch.

6.2 Basic Time Division Time Switching

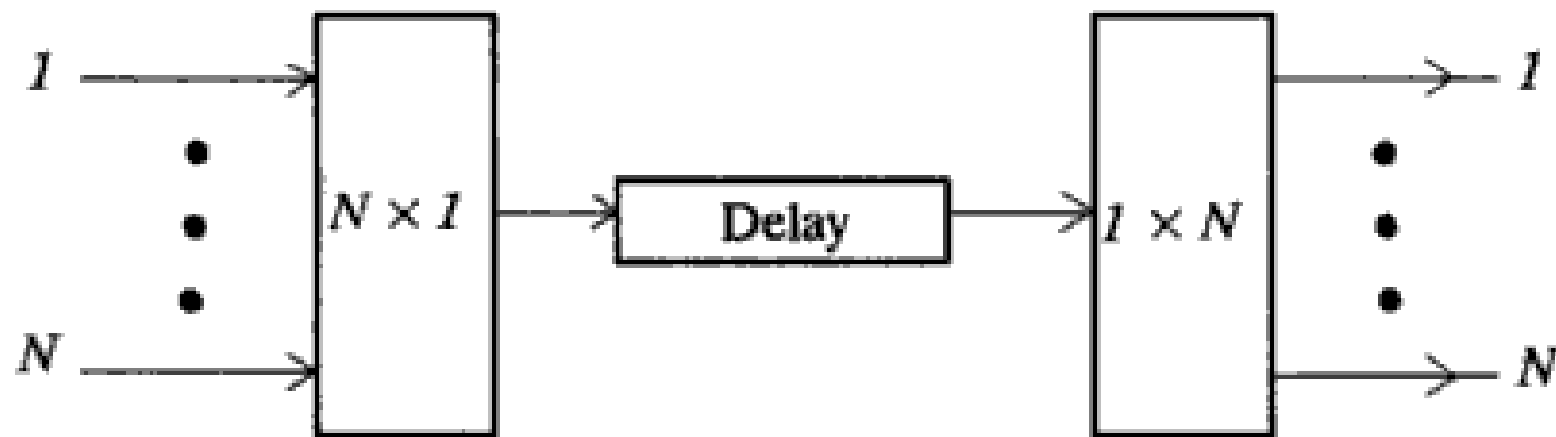
□ Switching Structure

□ Features

- Memory block is used instead of bus.
 - Suitable for PCM samples.
 - There is a time delay between acquisition of a sample from an inlet and its delivery to the corresponding outlet.
-



(a) Switching structure.



(b) Equivalent circuit.

Fig. 6.5 Basic time division time switching.

Switching Control Methods

- ☐ Sequential write/Random read
 - ☐ Random write/Sequential read
 - ☐ Random input/Random output
-

Switching Control Methods

- Sequential write/Random read
 - The inlets and outlets are both scanned sequentially.
 - The samples from inlets are written into data memory sequentially.
 - The samples are read from data memory randomly and then delivered to outlets.
-

Switching Control Methods

- Random write/Sequential read
 - The inlets and outlets are both scanned sequentially.
 - The samples from inlets are written into data memory randomly.
 - The samples are read from data memory sequentially and then delivered to outlets.
-

Switching Control Methods

- ❑ Radom input/Random output
 - The inlets and outlets are both scanned randomly.
 - Both data writing and data reading operations on data memory are performed sequentially.
-

Radom input/Random output

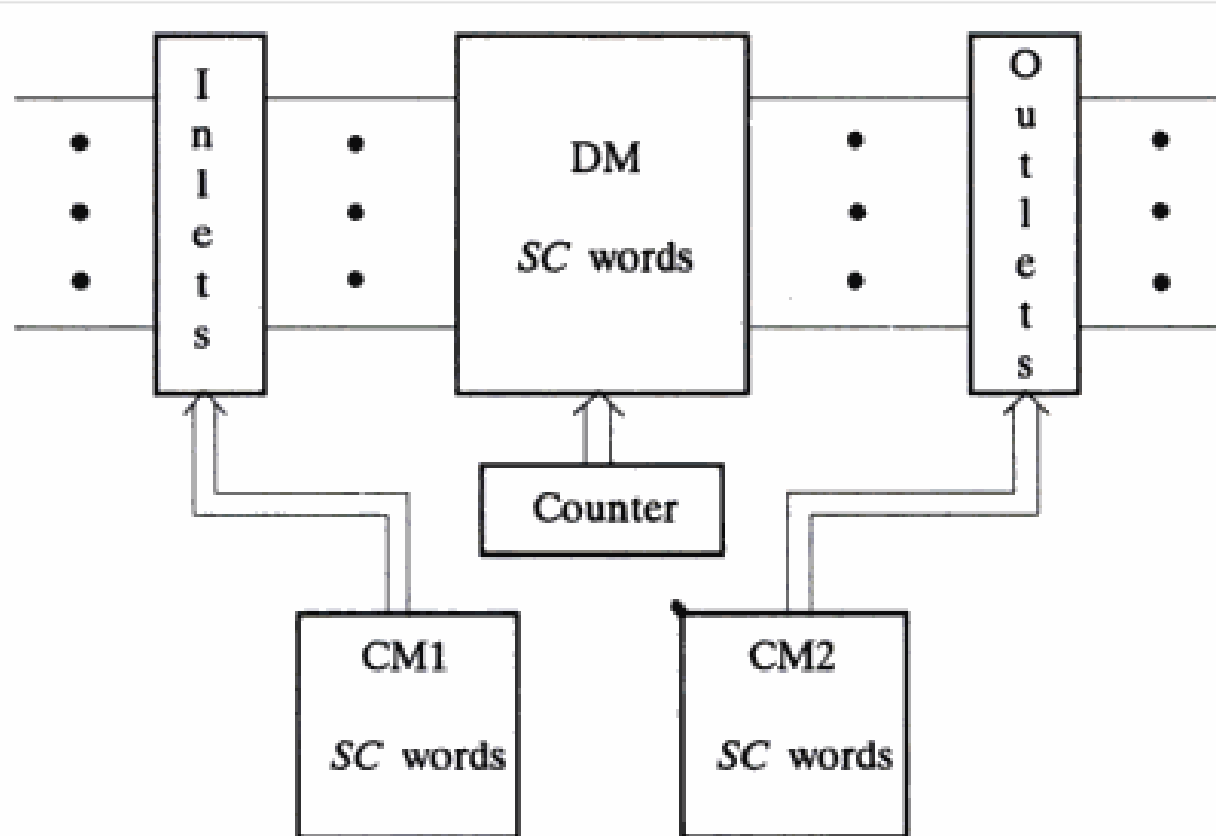


Fig. 6.8 Random input/random output time switch.

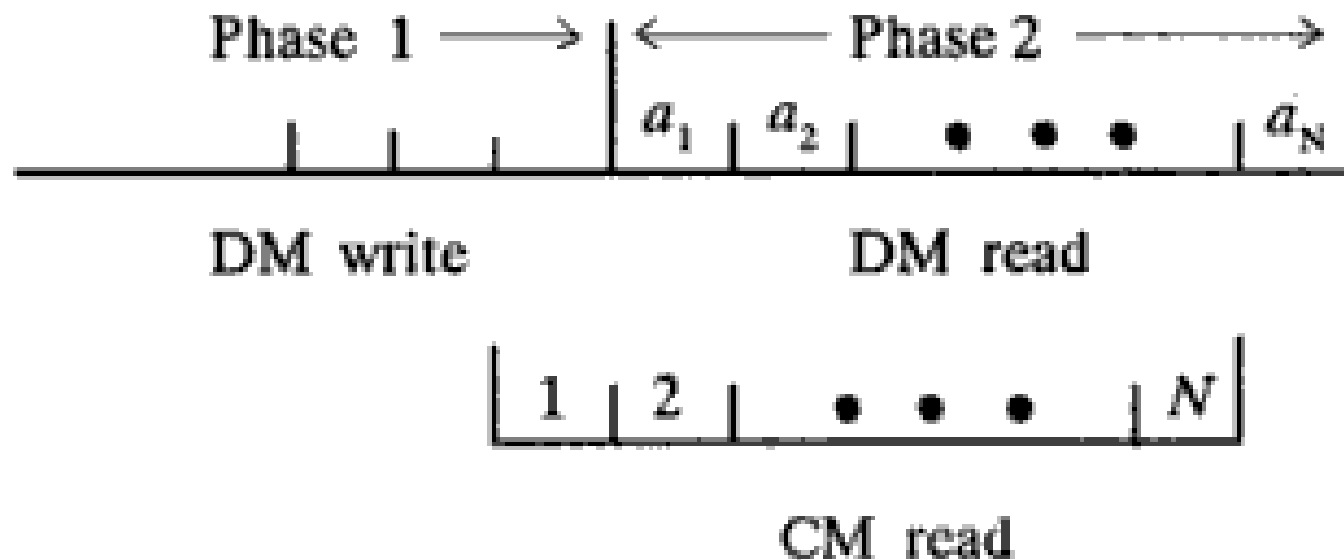
Operation modes

- ☐ Phased operation 阶段化操作模式
 - ☐ Slotted operation 时隙化操作模式
-

Operation modes

- Phased operation 阶段化操作模式
 - The switching procedure is divided into two phases.
 - Phase1: The data from inlets are stored in the data memory sequentially or randomly according to control method used.
 - Phase2: The data are read from data memory and delivered to outlets sequentially or randomly according to the control method used.
-

Phased operation

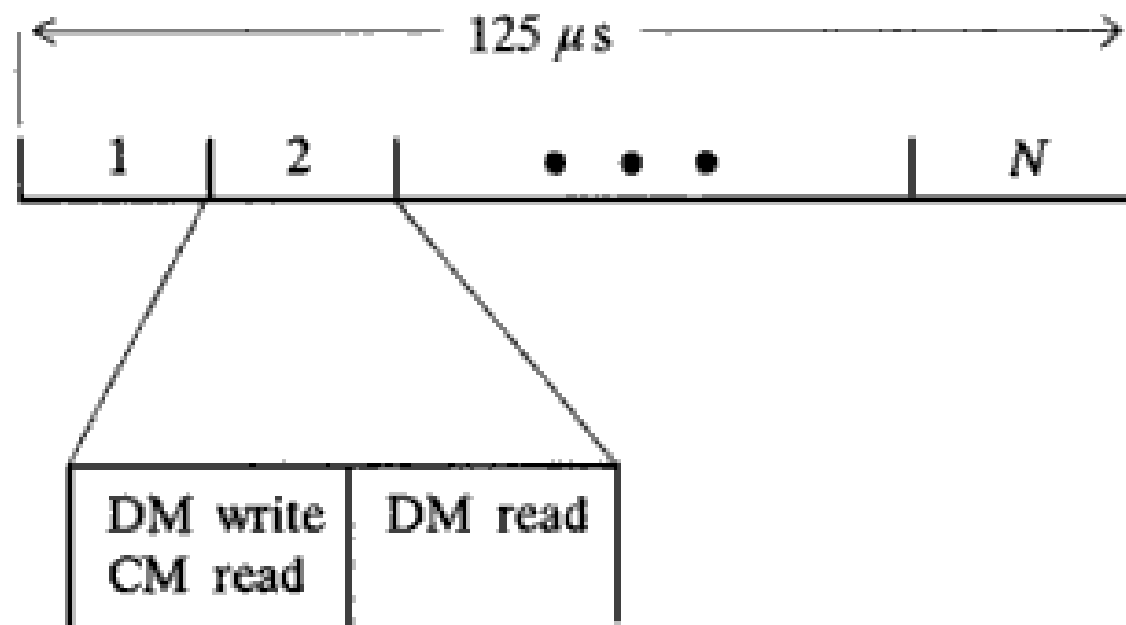


(a) Overlapping of data and control memory reads

Operation modes

- ❑ Slotted operation 时隙化操作模式
 - The operation period (125us) is divided into slots according to switching capacity.
 - In each time slot, the data from a inlet is stored in data memory, and then a data is read and delivered to its corresponding outlet.
 - One sample delay (125us) may be introduced.
-

Slotted operation



(b) Simultaneous DM write and CM read

Fig. 6.6 Two modes of operation of time switches.

Remarks

- ❑ Both sequential write/random read and random write/sequential read control modes are **nonblocking** in nature, but the **number of subscribers** can be connected to the system is **no more than the switching capacity SC**.
 - ❑ Random input/Random output control mode permits a **large number of subscribers** connected to the system, but it is **blocking** in nature.
 - ❑ Each of the inlets/outlets corresponds to a single subscriber. **Suitable for local exchanges**.
-

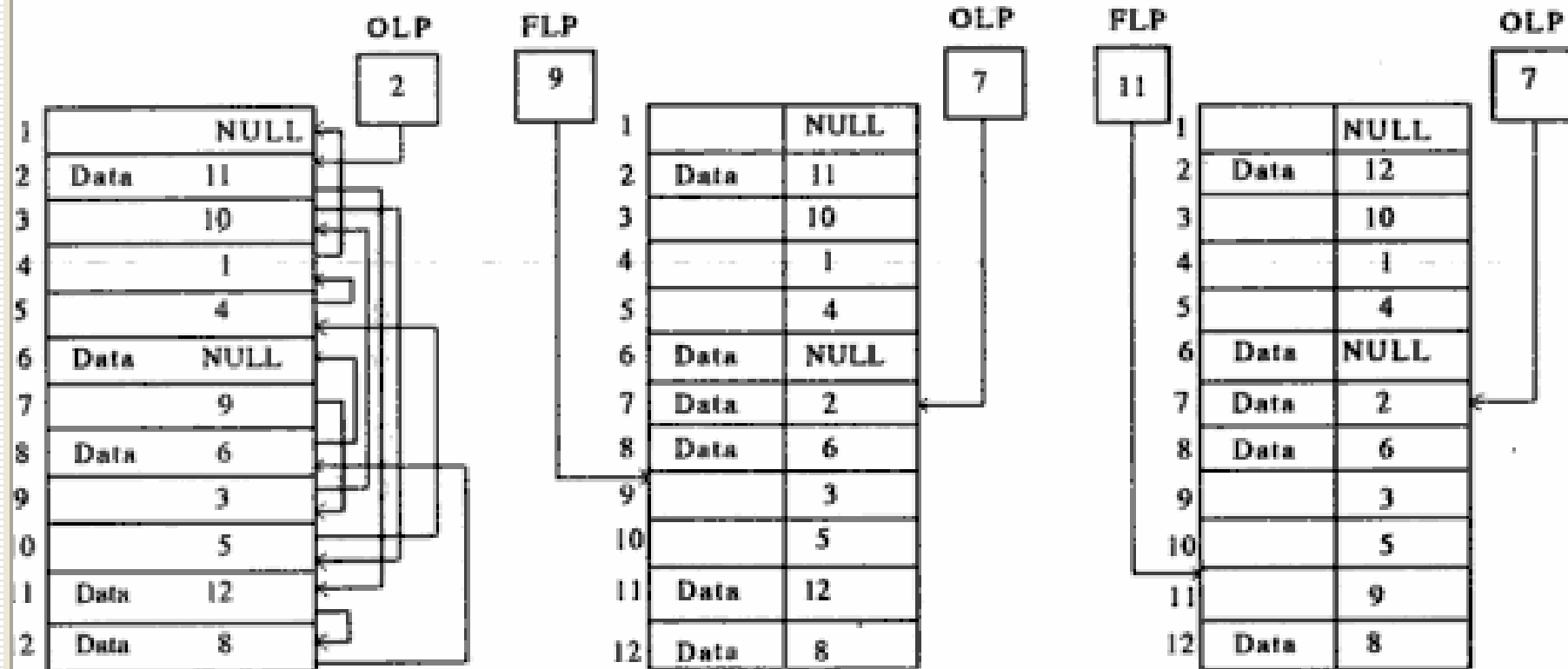
Control memory management in random input/random output control

□ Control memory configuration

- A number of memories (CM1/CM2) are shared by all inlets and outlets.
- How to share and manage the control memory efficiently?

□ Management methods

- Maintaining a free list
 - Compacting the entries every time a call terminate
 - Maintaining free and occupied lists
-



FLP = free list pointer OLP = occupied list pointer

(a) A sample list

(b) Location 7 allocated

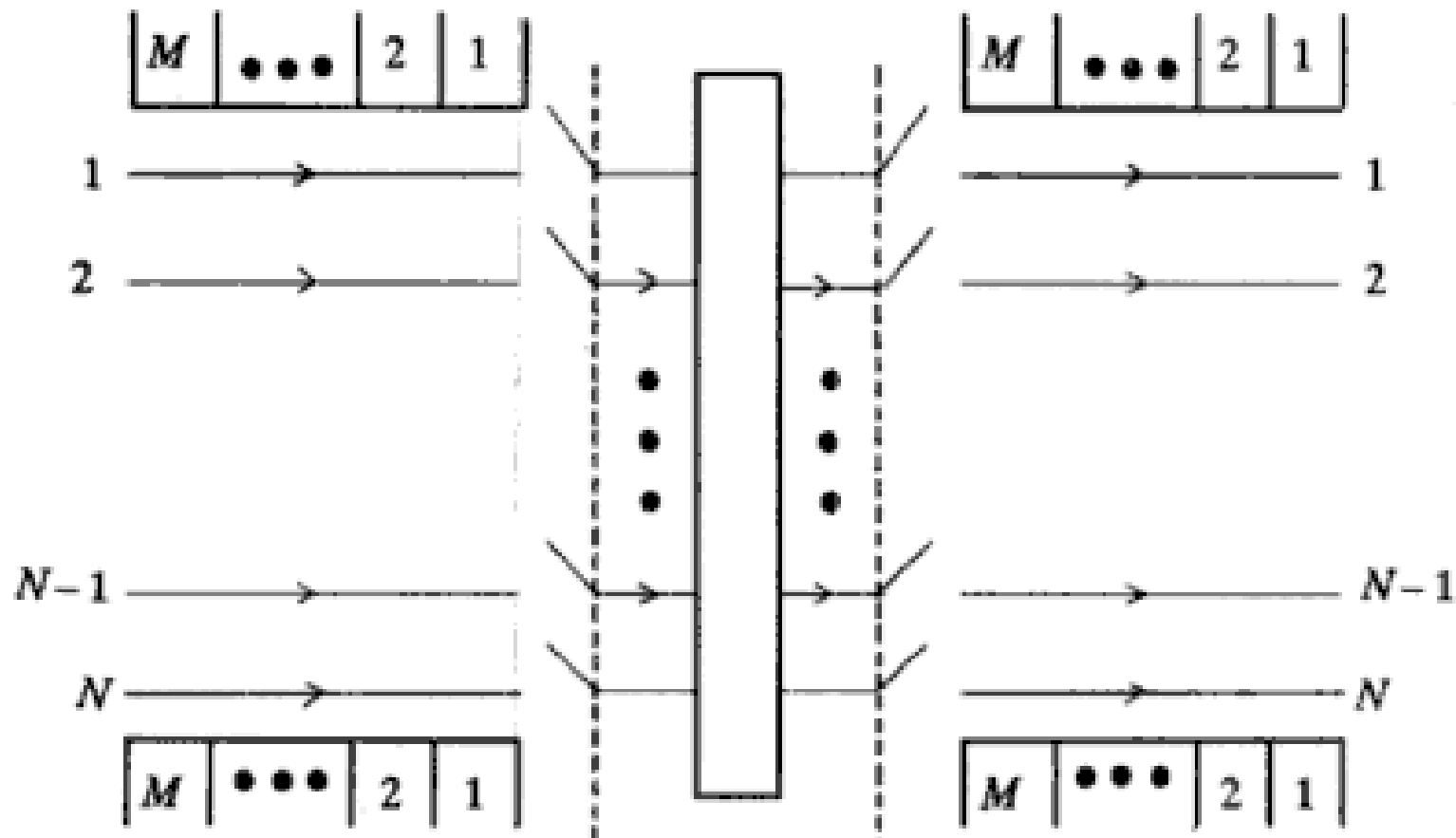
(c) Location 11 freed

Fig. 6.9 List management.

6.3 Time multiplexed Space Switching

- In transit **exchanges**, the inlets and outlets are **time division multiplexed trunks**.
 - How to realize Switching in transit exchanges?
 - Time multiplexed switches
 - Time multiplexed **Space Switches**
 - Time multiplexed **Time Switches**
 - **Combination Switches**
-

Time multiplexed switching



Time Multiplexed Space Switching

□ Configuration Parameters

- N incoming trunks & N outgoing trunks
 - M samples per frame
 - Frame duration: 125us
 - Time slot duration: $125/M$ us
 - Number of speech samples switched in one frame time: NM
 - Number of speech samples switched in one time slot: N
-

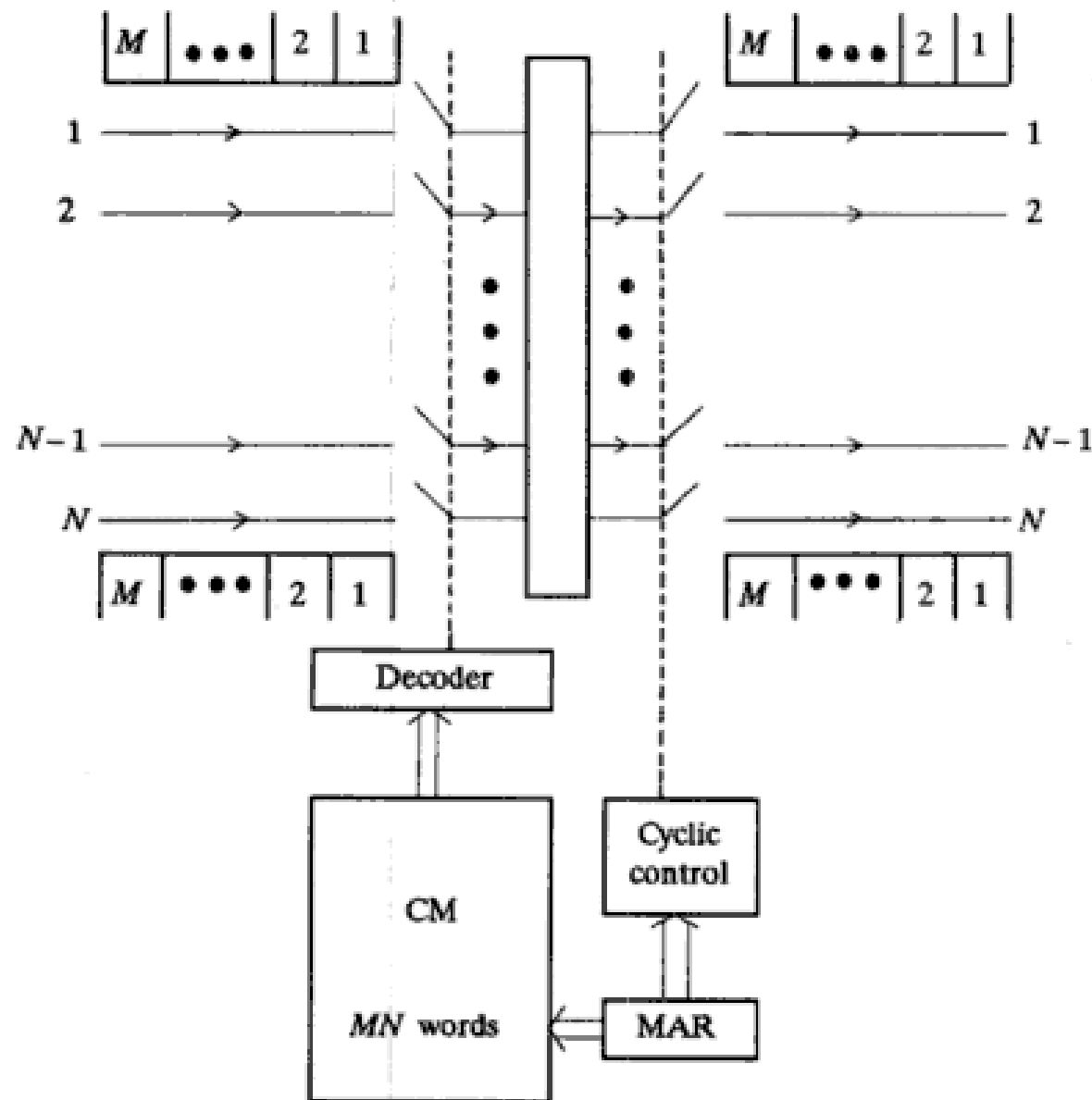


Fig. 6.10 Time multiplexed space switch.

Output-controlled configuration

CM: control memory
 t_s : switching time per Inlet-outlet pair.

Time Multiplexed Space Switching

☐ Principle

- The outgoing trunks are scanned cyclically.
- The incoming trunks are controlled by CM.

☐ Number of trunks supported

- $N = 125 / (M t_s)$
- How to improve N?

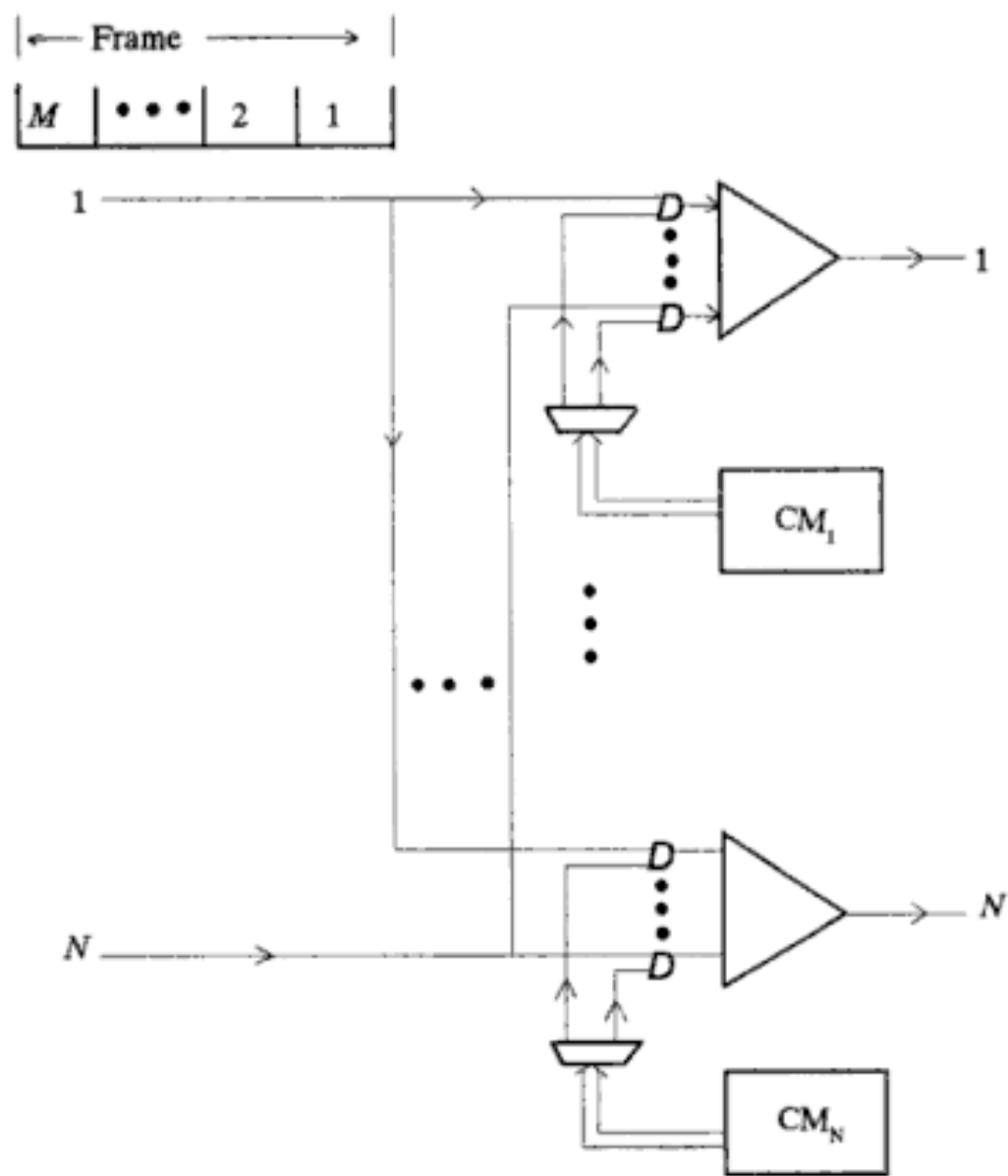
☐ Cost

- $C = \text{No. of switches} + \text{No. of memry words}$
 $= 2N + MN$

☐ Equivalent Input-controlled scheme can be implemented similarly.

A modified scheme

- How to improve the number of trunks supported?
 - Key point: time cost of CM access
 - A feasible solution:
 - One CM for each output line
 - Parallel CM access
 - There is no constraint on N due CM access time.
 - Cost: $C = N \times N + NM = N^2 + NM$
 - Much more expensive than the former.
-



CM = control memory

Fig. 6.11 Time multiplexed space switch with N control memory modules.

Remarks

- ☐ Word width: $\log_2 N$
 - ☐ Full availability: no
 - Subscribers belonging to different time slots can not be connected.
 - ☐ Broadcast: yes
-

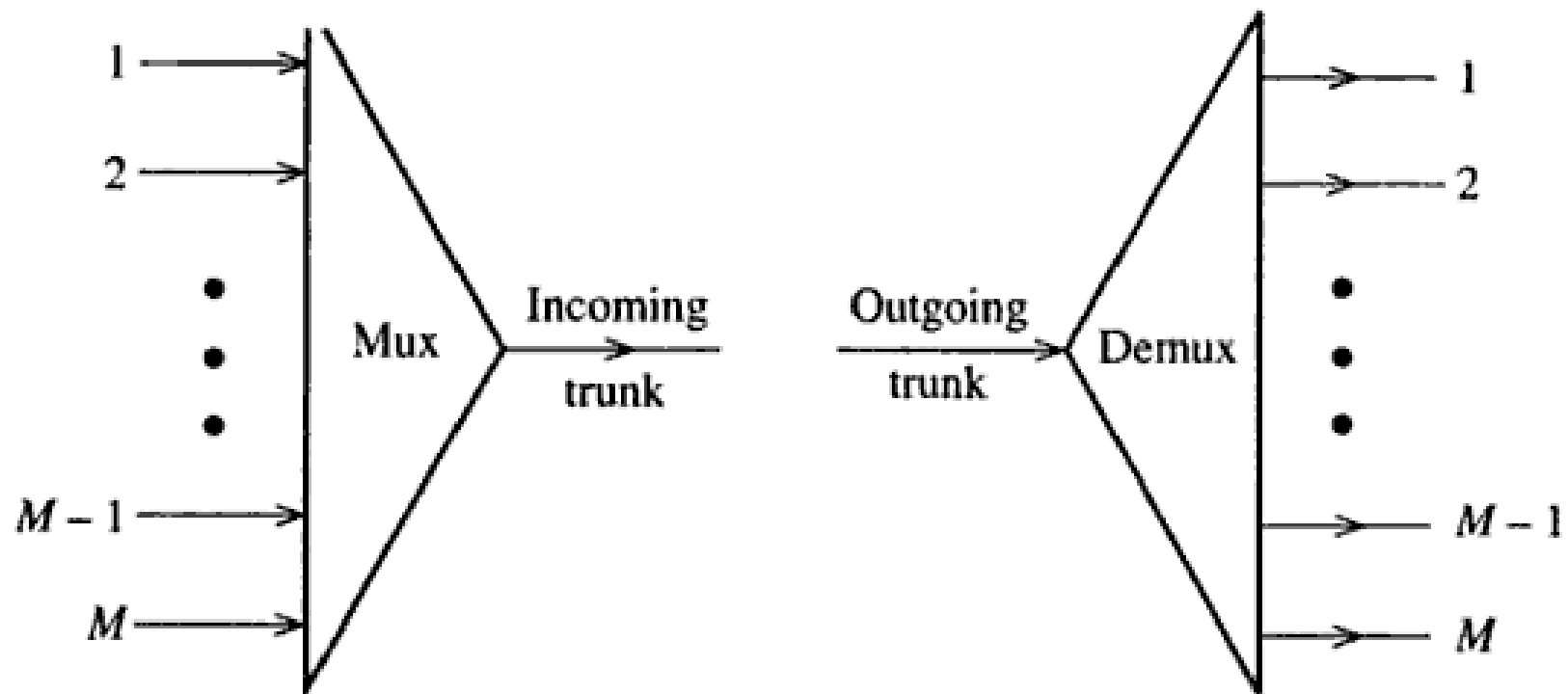


Fig. 6.12 Source multiplexing and destination demultiplexing.

A realization based on space array

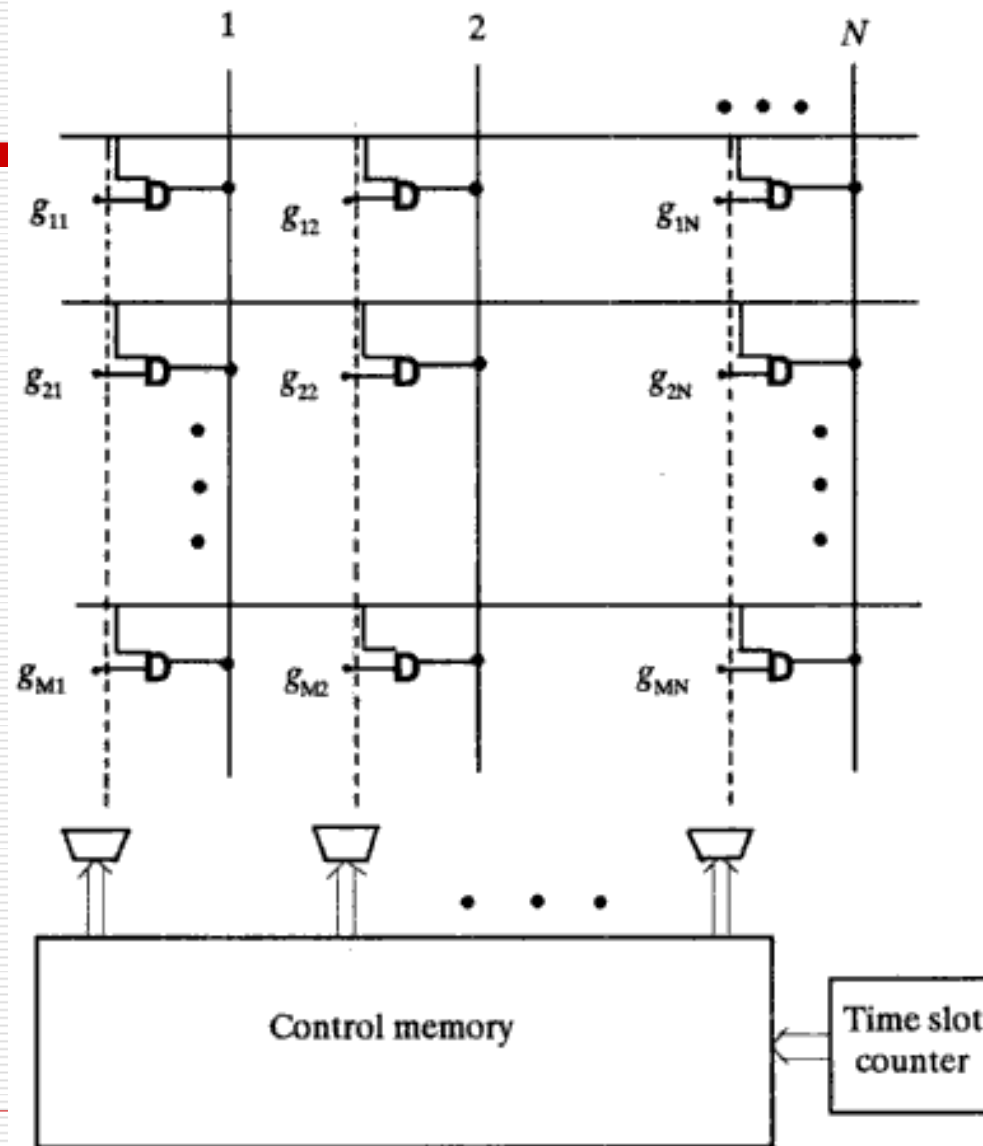
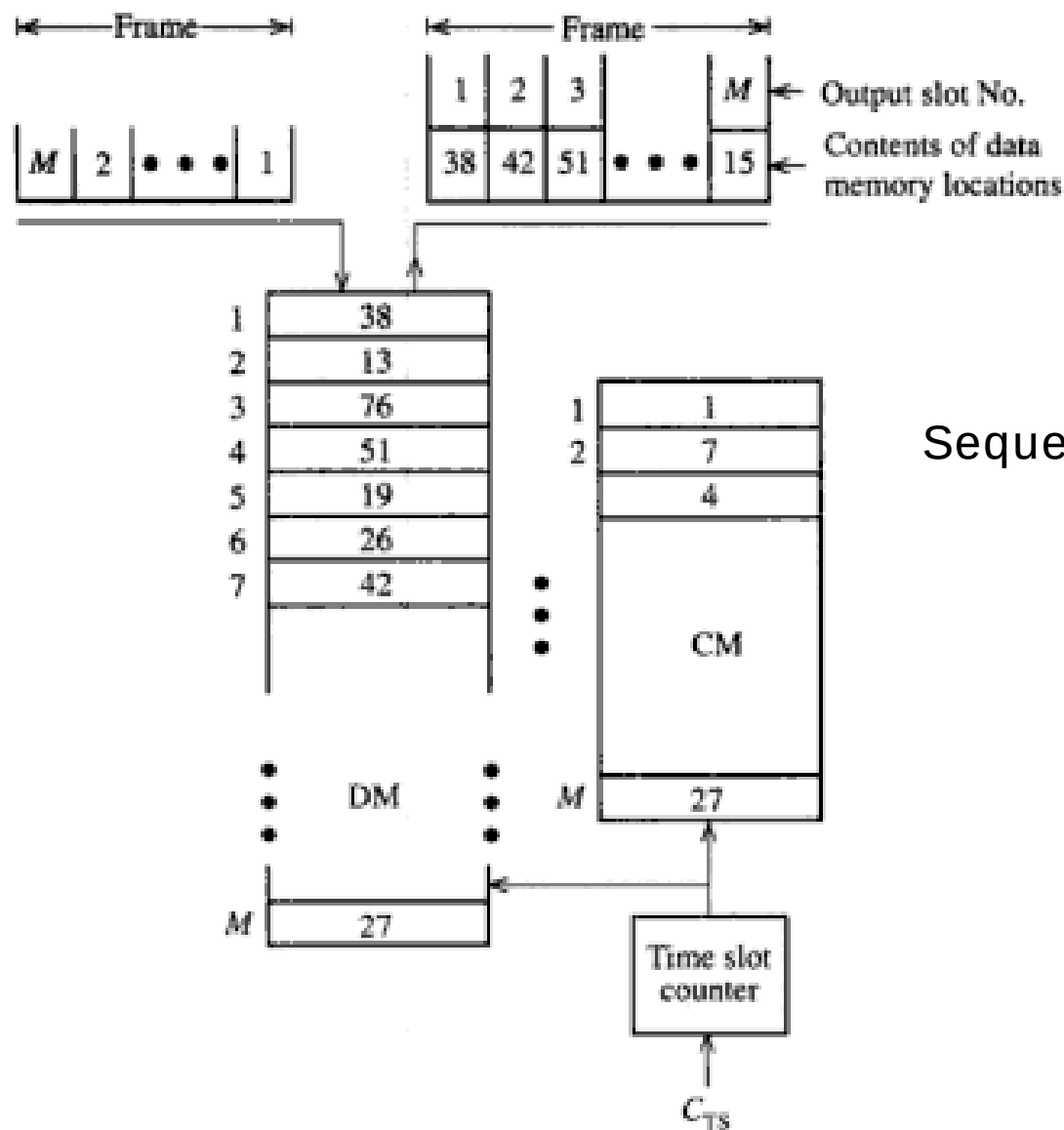


Fig. 6.13 Time division space switch using space array.

6.4 Time multiplexed time switching

- How to realize switching among subscribers belonging to different time slots?
 - Memory write/read based method
 - Principle of time slot interchange (TSI)
 - Category:
 - Random write/Sequential read
 - Sequential write/Random read
 - Sample time delays
-

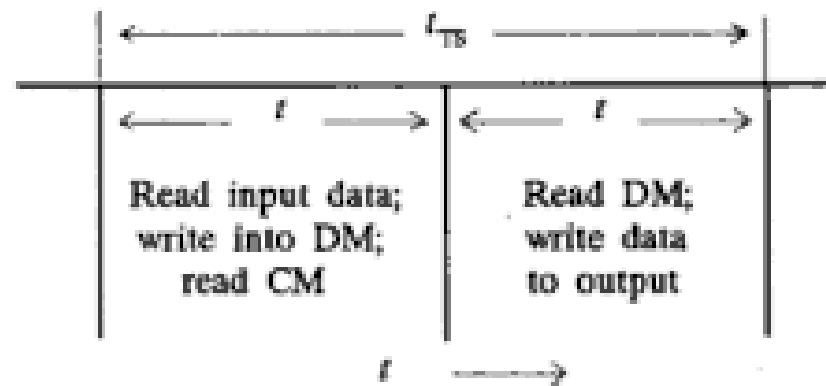


Sequential write/Random read

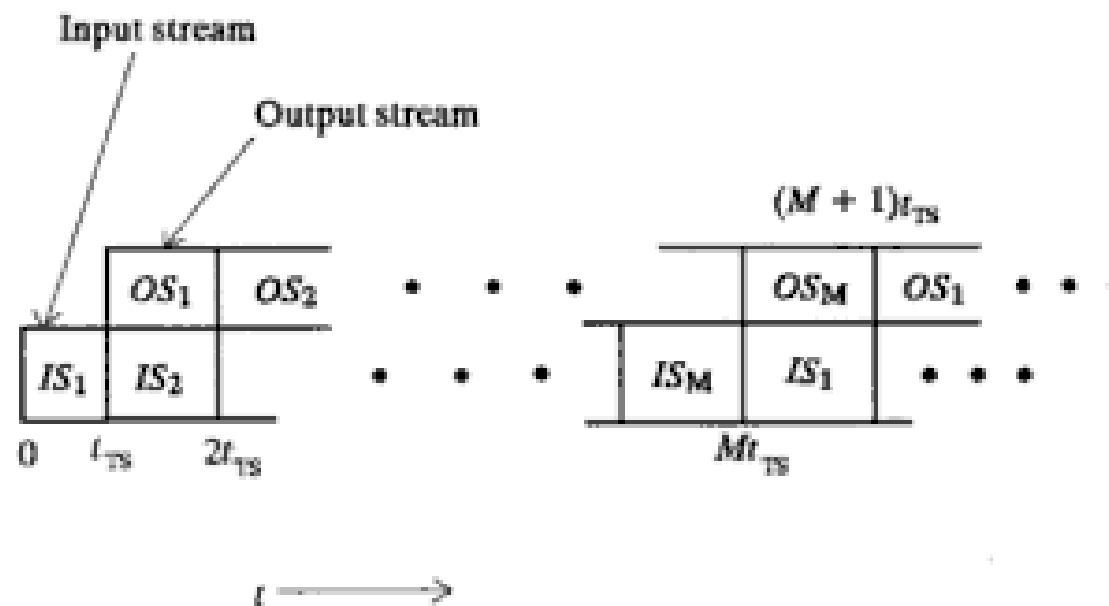
CM = control memory DTS = time slot clock DM = data memory

(a) Principle of time slot interchange

Fig. 6.14 Time slot interchange switch.



(b) Operations in a time slot



IS = input slot OS = output slot

(c) Time relationship between inlet and outlet streams

Fig. 6.14 Time slot interchange switch (cont.).

Principle of time slot interchange (TSI)

- Frame duration: 125us
 - Time slot duration: $t_{Ts} = 125/M$ us
 - Time constraints:
 - $t_{Ts} = 2t_m$
 - $125 = 2Mt_m$
 - t_m : access time of memory.
 - Cost:
 - $C = \text{No. of Switch} + \text{No. of Memory words}$
 $= 0 + 2M = 2M$
-

Expanding & Concentrating in TSI

□ Expanding

- The number of time slots per frame in the output stream $M2$ is larger than that in the input stream $M1$, i.e. $M2 > M1$.

□ Concentrating

- The number of time slots per frame in the input stream $M1$ is larger than that in the output stream $M2$, i.e. $M1 > M2$.

□ Implementation:

Independent/asynchronous write and read.

□ Constraint: $125 = (M1 + M2)t_m$

Expanding TSI

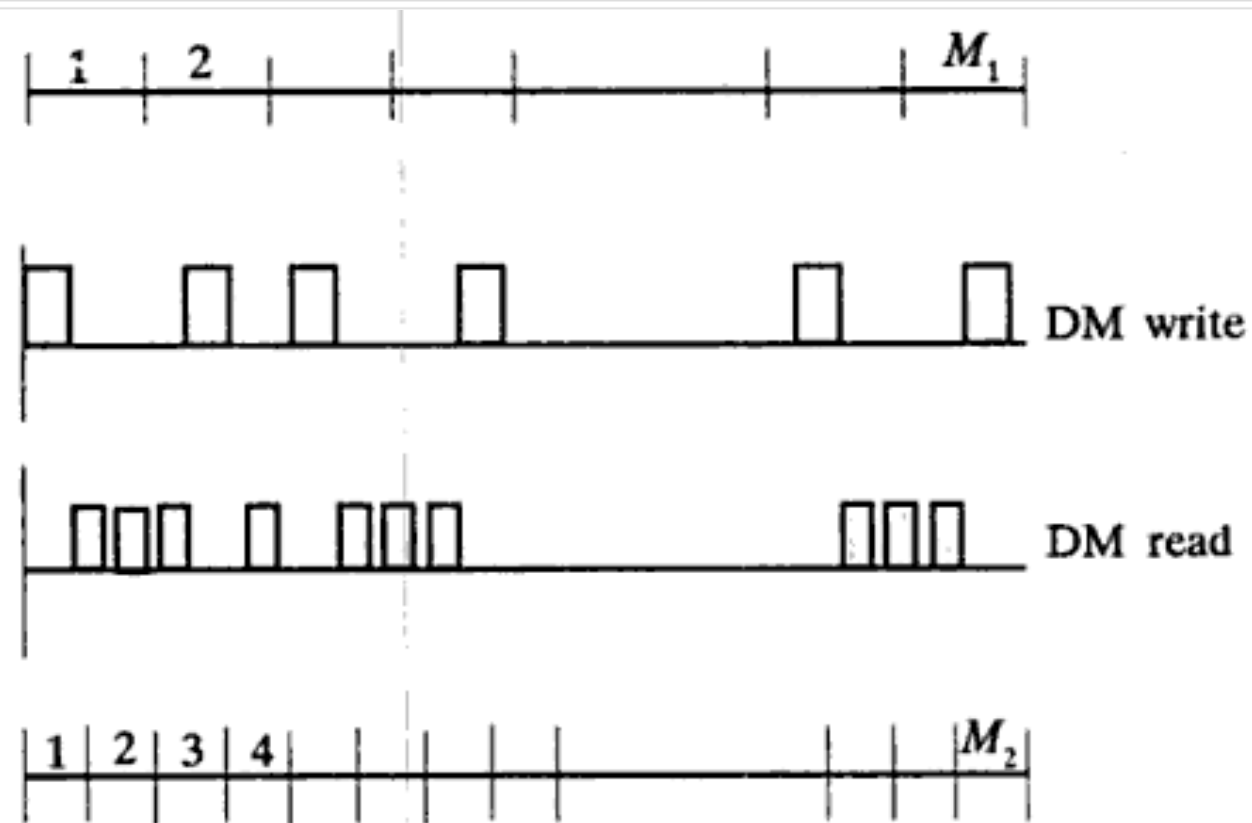


Fig. 6.15 Asynchronous read/write operations.

Time multiplexed time switching

□ Problem faced

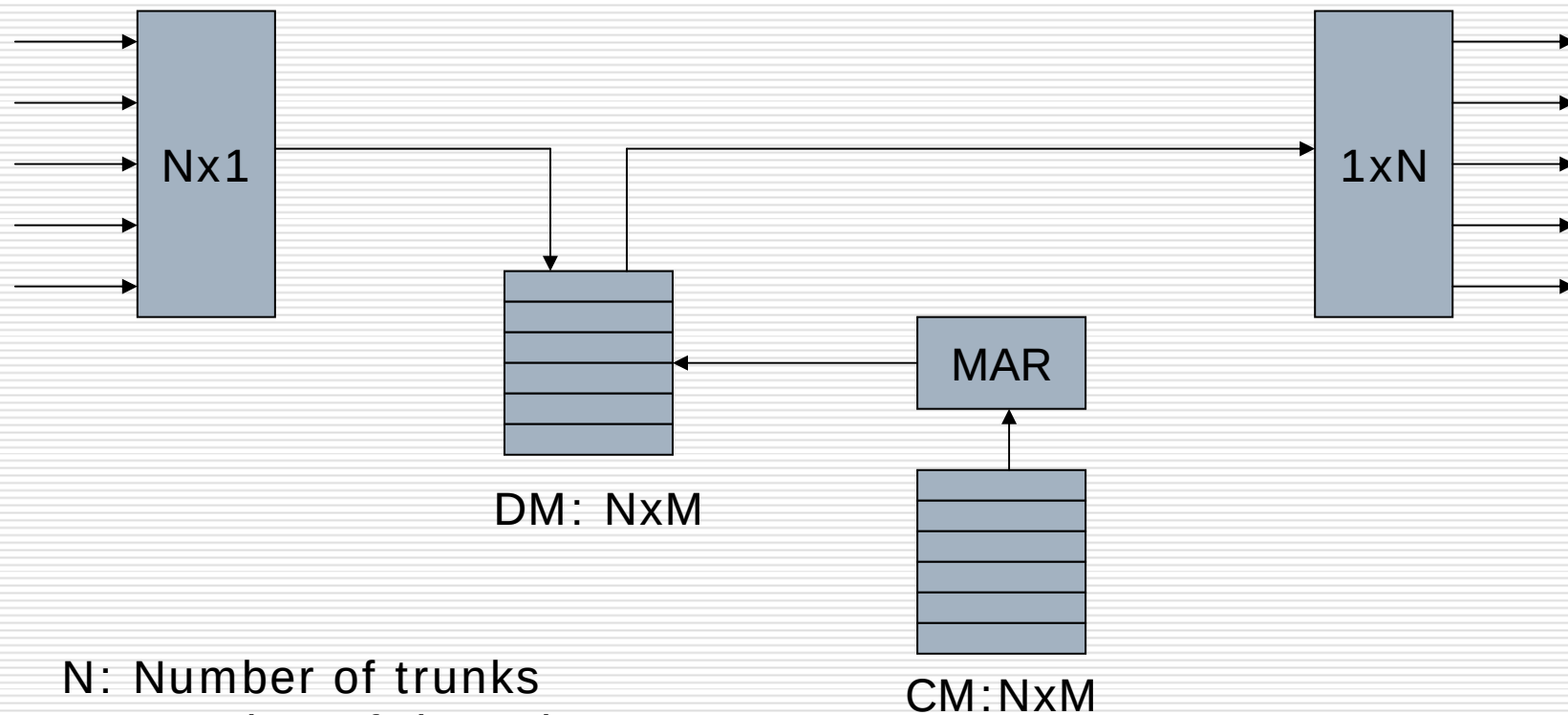
- N time multiplexed input streams each multiplexing M subscribers
 - N time multiplexed output streams each carrying M subscribers
 - The problem is to handle NM subscribers in the time duration of 125us.
-

Time multiplexed time switching

□ Practical Configurations

- Serial-in/Serial-out (串入串出)
 - Parallel-in/Serial-out (并入串出)
 - Serial-in/Parallel-out (串入并出)
 - Parallel-in/Parallel-out (并入并出)
-

Serial-in/Serial-out (串入串出)

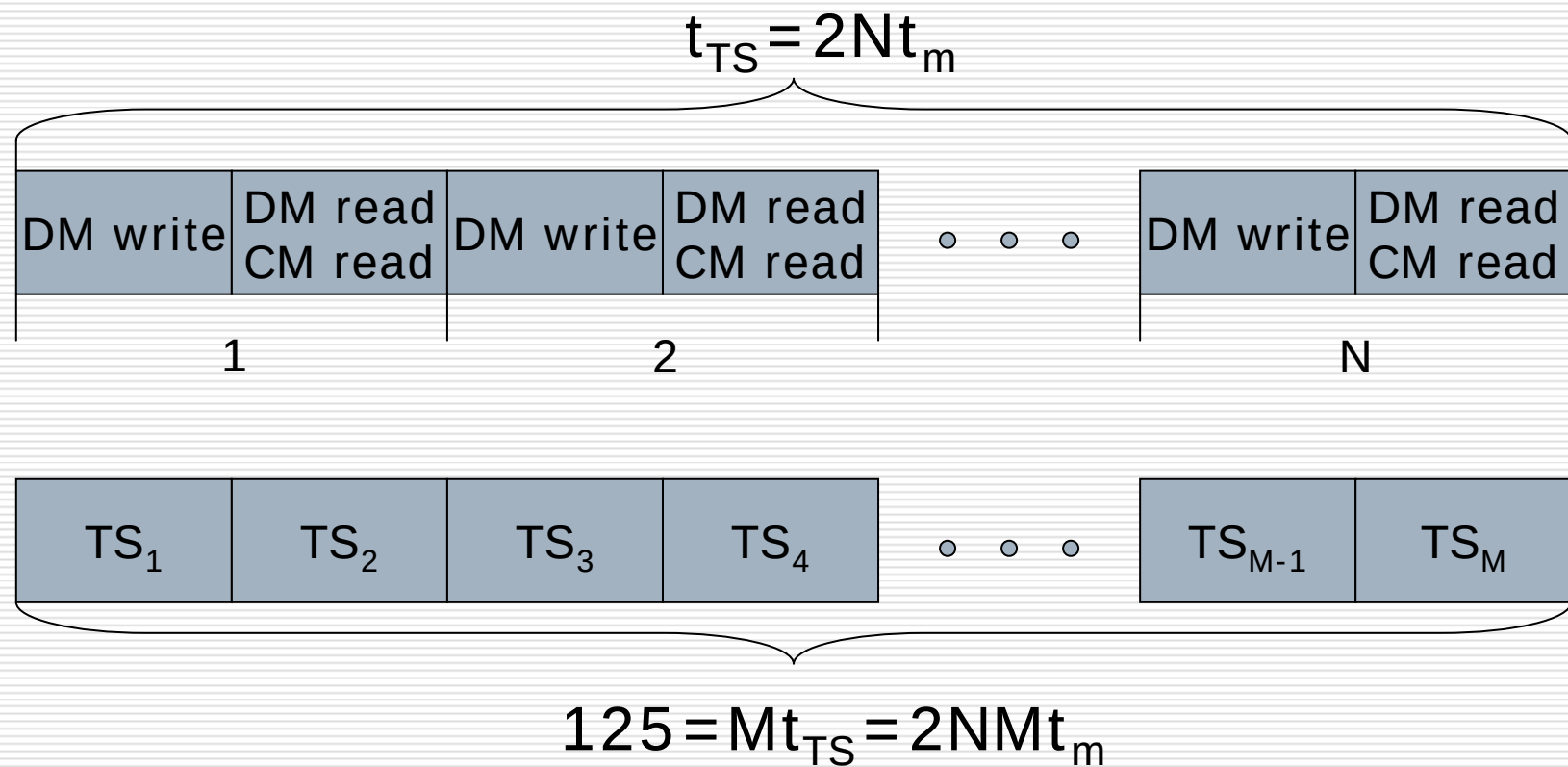


N : Number of trunks

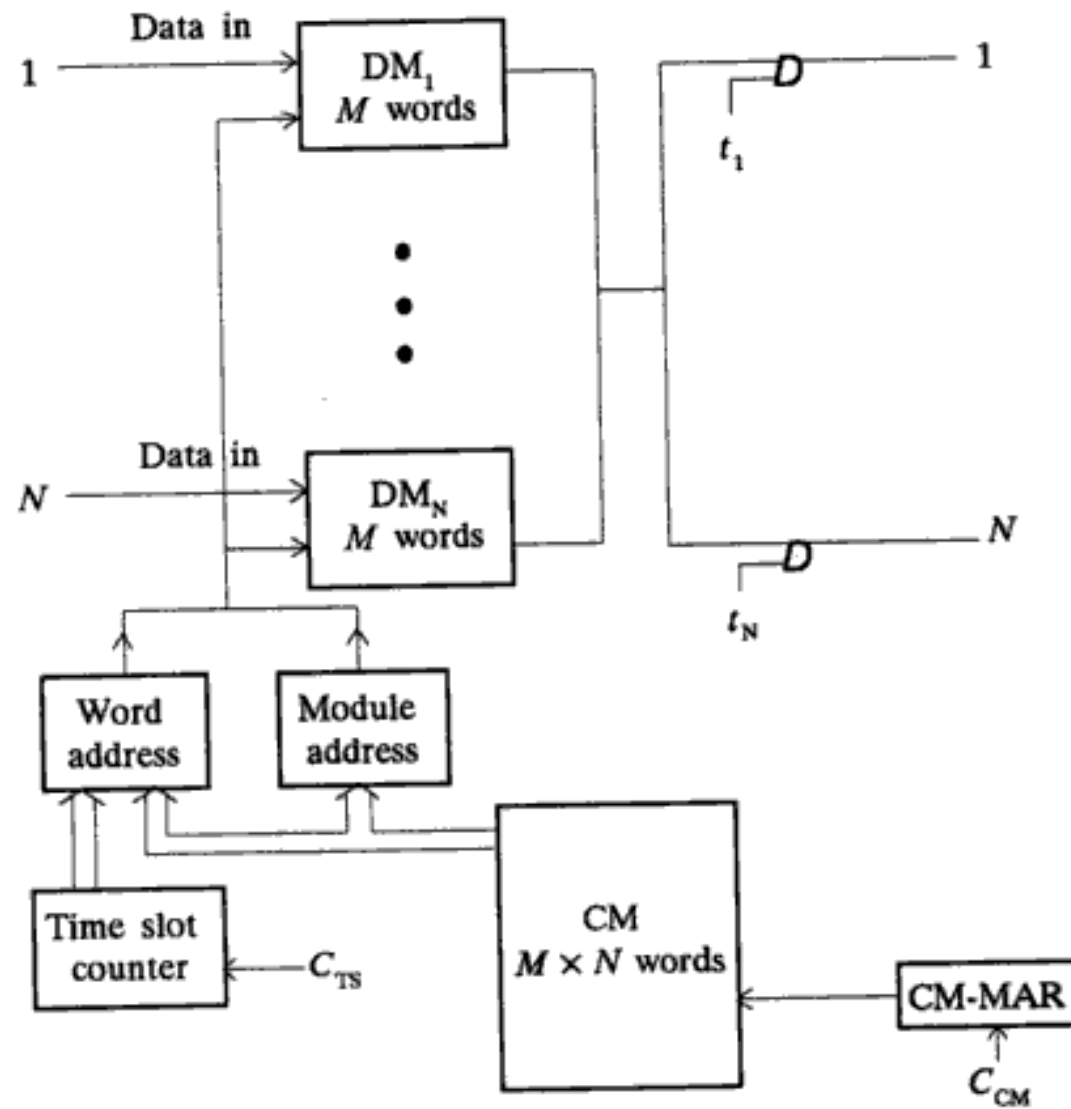
M : Number of time slots

Time constraints: $t_{TS} = 2Nt_m$ $125 = 2NMt_m$

Serial-in/Serial-out (串入串出)

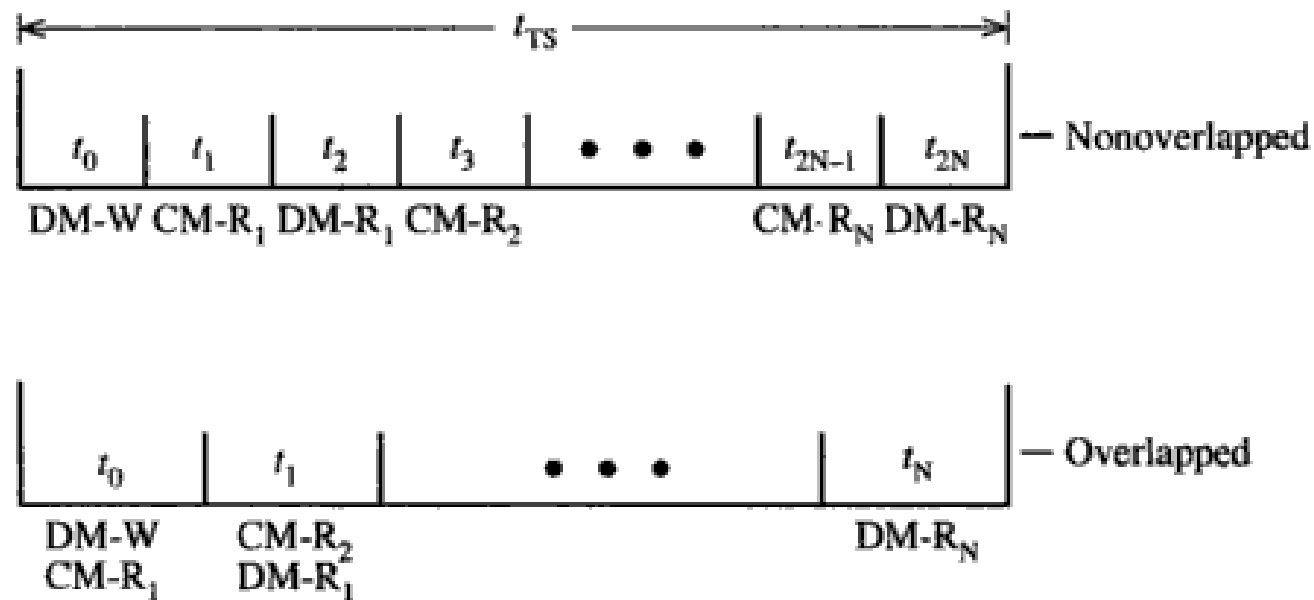


Parallel-in/Serial-out (并入串出)



(a) Configuration

Parallel-in/Serial-out (并入串出)

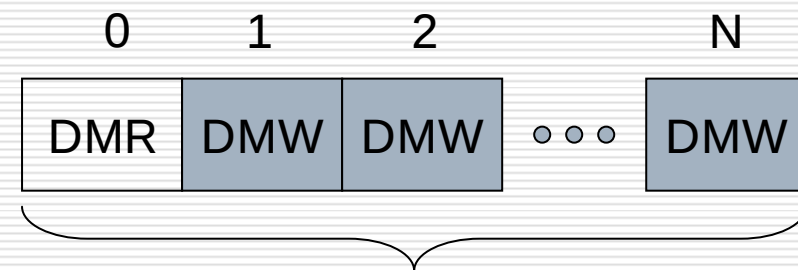
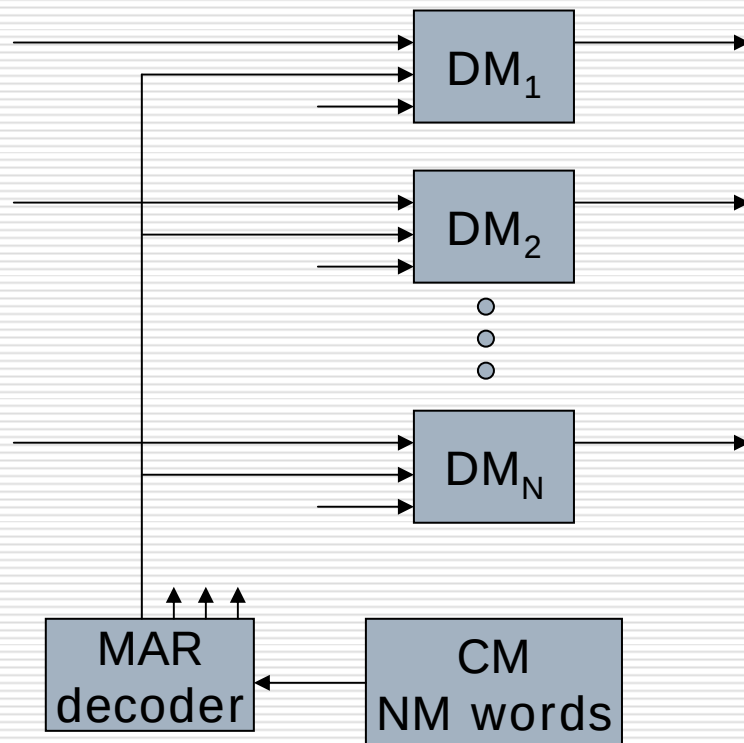


(b) Timing details

Fig. 6.16 Parallel-in/serial-out configuration.

Time constraints: $t_{TS} = (2N+1)t_m$ $125 = 2NMt_m$

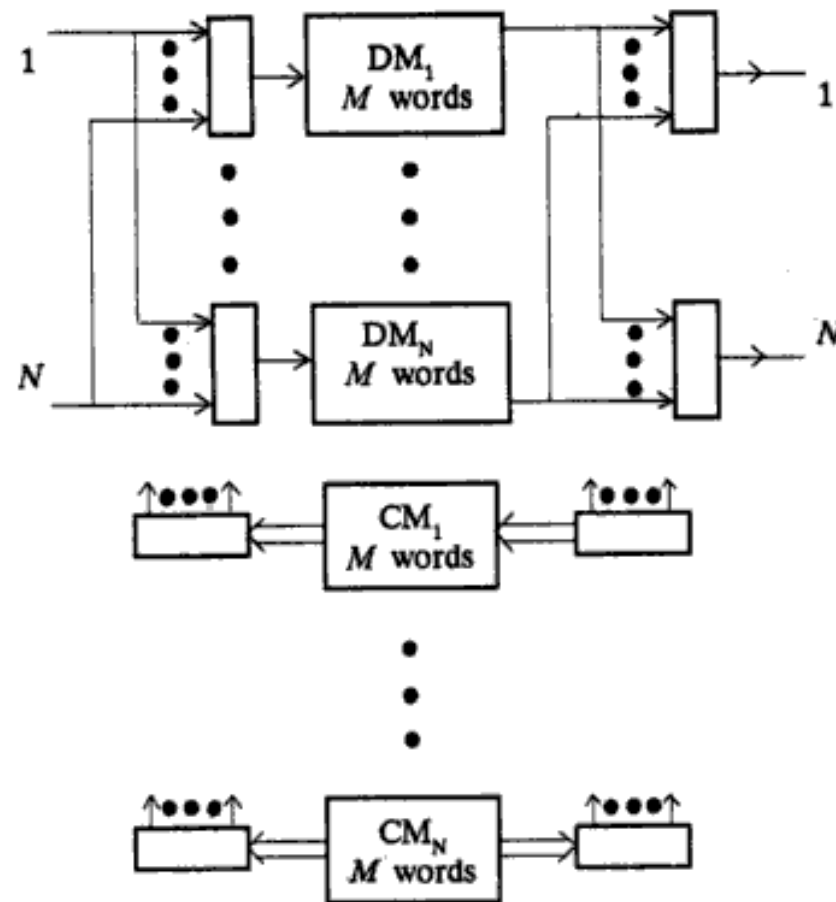
Serial-in/Parallel-out (串入并出)



$$t_{TS} = 125/M = (2N+1)t_m$$

$$125 = Mt_{TS} = (2N+1)Mt_m$$

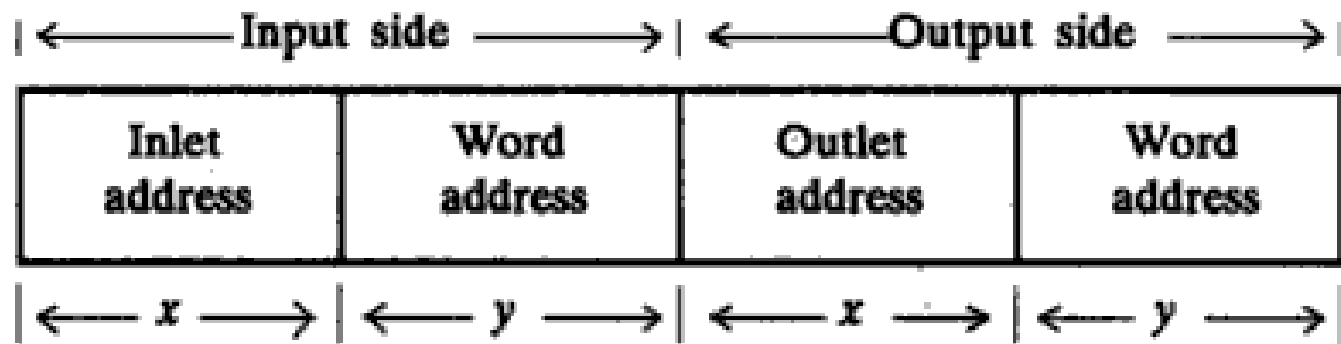
Parallel-in/Parallel-out (并入并出)



CM = control memory DM = data memory

(a) Configuration

Parallel-in/Parallel-out (并入并出)



$$x = \lceil \log_2 N \rceil \quad y = \lceil \log_2 M \rceil$$

(b) Contents of control memory locations

Fig. 6.17 Parallel-in/parallel-out time switch.

Remarks

- ❑ Time multiplexed time switches do not provide full availability, because they are not capable of switching samples across trunks.
-

6.5 Combination Switching

- How to provide full availability for time multiplexed trunks?
 - Both space switching and time slot switching should be performed.
 - Combination Switching
 - Basic idea: Multistage & Space + Time
 - Category
 - Two-stage
 - Three-stage
 - multistage
-

Two-stage combination switches

□ Configurations of two-stage ~

■ Time-space (TS) switch

- The first stage performs time switching;
- The second stage performs space switching.

■ Space-time (ST) switch

- The first stage performs space switching;
 - The second stage performs time switching.
-

Time-space (TS) switch

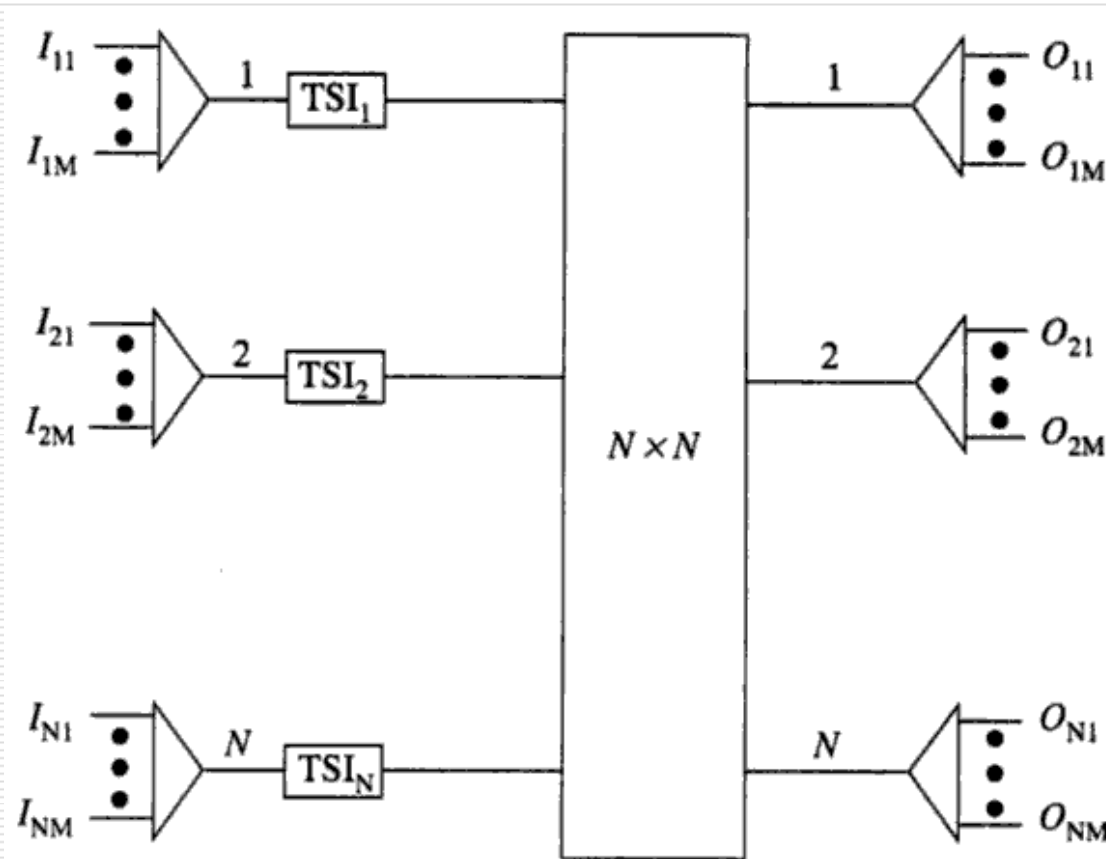


Fig. 6.18 Two-stage TS switch.

Example

Connection:

$I_{47} \longrightarrow O_{56}$

Stage 1:

$I_{47} \longrightarrow I_{46}$

Stage 2:

$I_{46} \longrightarrow O_{56}$

Remarks

- ❑ This two-stage TS switch ensures **full availability**.
- ❑ It is a **blocking** network. If two or more samples belonging to a specific inlet are destined to the same time slot in different outlets, **blocking** will occur.



Space-time (ST) switch

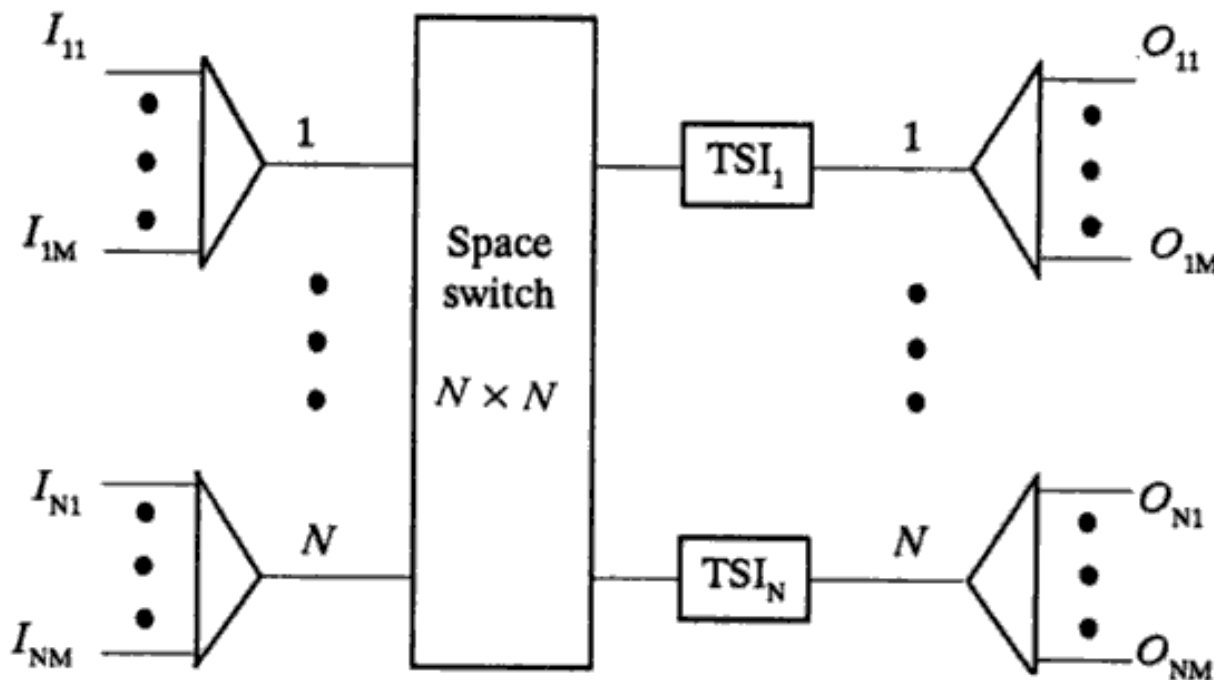


Fig. 6.19 A space-time switch.

Example

Connection:

$I_{47} \longrightarrow O_{56}$

Stage 1:

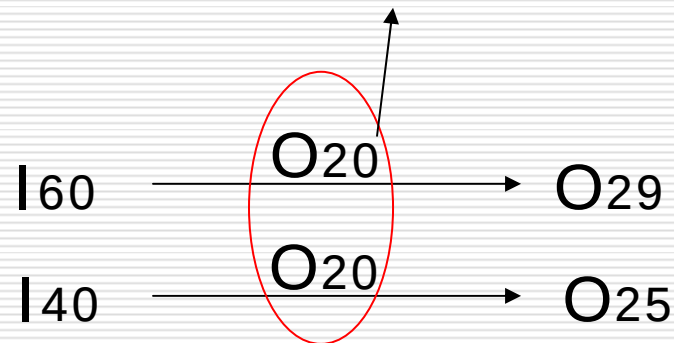
$I_{47} \longrightarrow O_{57}$

Stage 2:

$O_{57} \longrightarrow O_{56}$

Remarks

- ❑ This two-stage ST switch ensures **full availability**.
- ❑ It is a **blocking** network. If two or more samples originating from different inlets during the same time slot are destined to the same outlet, **blocking** will occur.



6.6 Three-stage Combination Switching

□ Category

- Time-space-time (TST) switches
 - The first and third stages perform time switching;
 - The second stage performs space switching.
 - Space-time-space (STS) switches
 - The first and third stages perform space switching;
 - The second stage performs time switching.
 - Both TST and STS are blocking in general cases.
-

Time-space-time (TST) switching

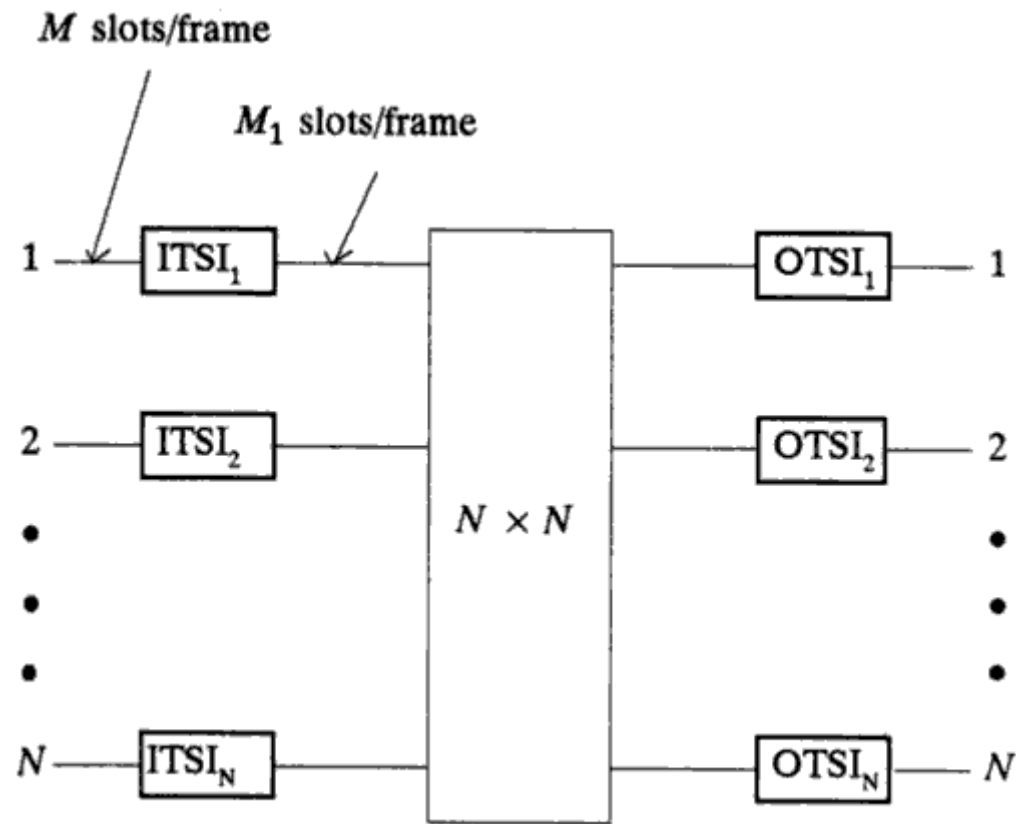


Fig. 6.20 A time-space-time switch.

Space-time-space (STS) switching

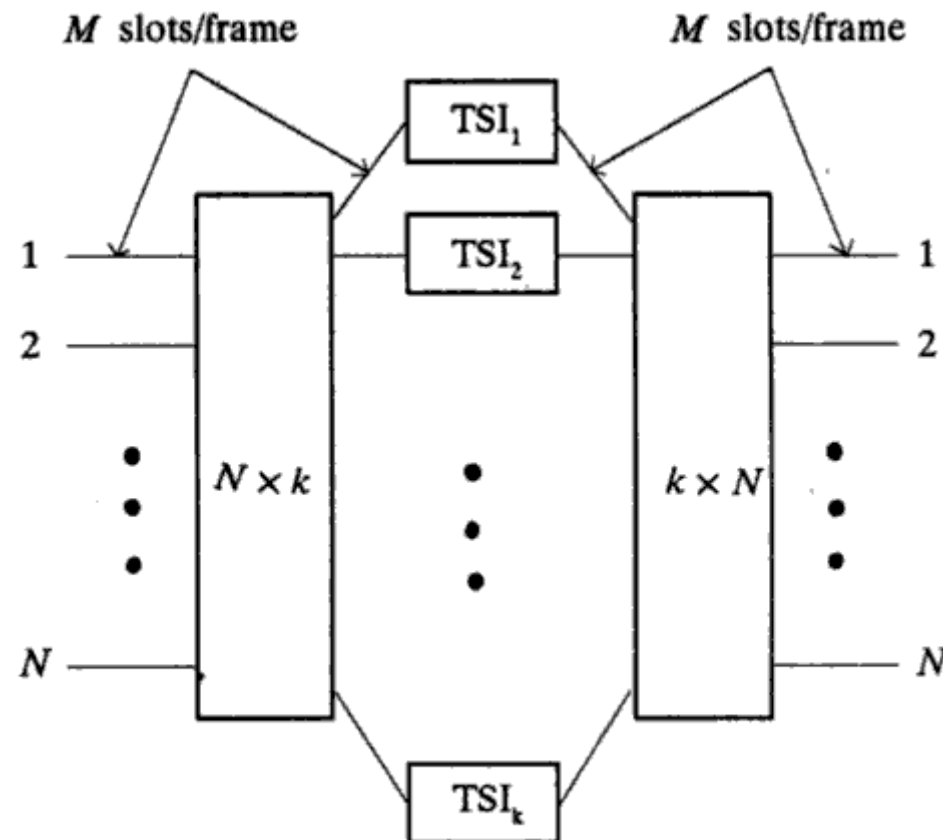


Fig. 6.22 A STS switch.

Costs of TST and STS

□ TST

$$C_{\text{TST}} = 5MN + N^2$$

□ STS

$$C_{\text{STS}} = 2Nk + 4MN$$

- The blocking probability is reduced by providing more feasible paths for any inlet-outlet pair.
-

6.7 n-Stage Combination Switching

Table 6.1 Examples of Time-Space Combination Switches

System	Configuration	Traffic capacity (erlangs)	Maximum number of trunks
E10 B (France)	TST	1600	3600
No.4 ESS (USA)	TSSSST	47,000	107,520
C-DOT MAX-XL (India)	TST	16,000	40,000
DMS 100 (Folded) (Canada)	TSTS	39,000	61,000
System 12 (USA)	TSTSTSTSTSTSTS	25,000	60,000
NEAX 61 (Japan)	TSST	22,000	60,000

Assignments

☐ Ex.2

☐ Ex.6
